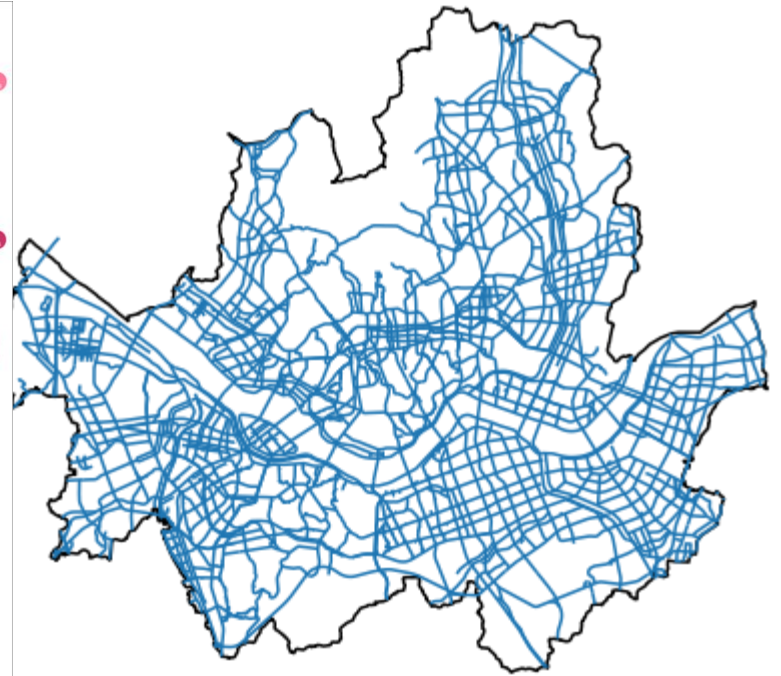
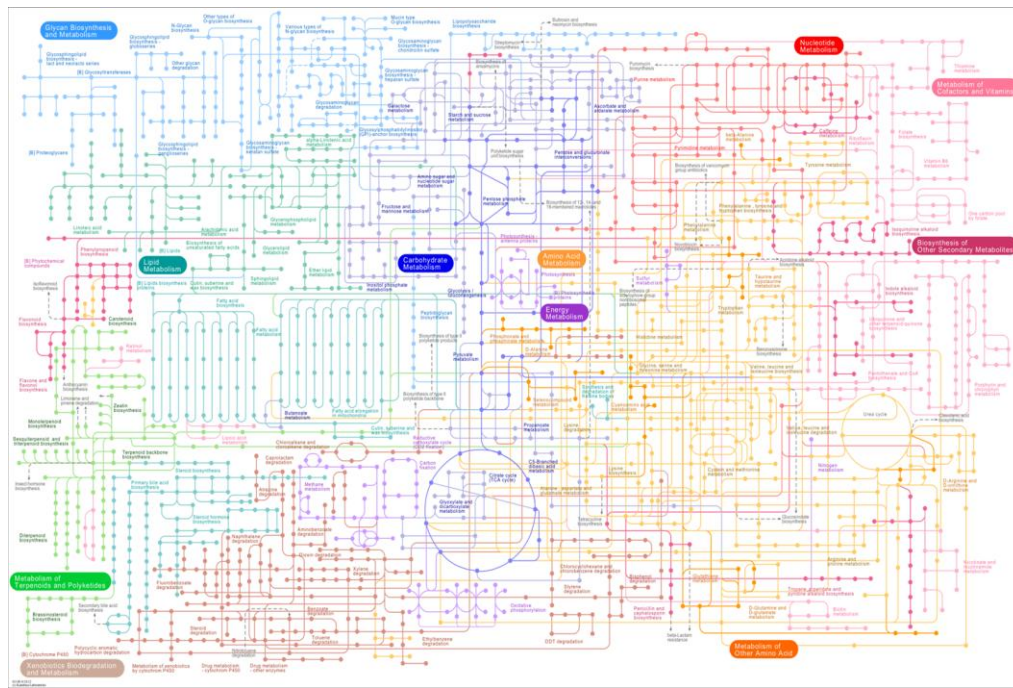




David L. Nelson and Michael M. Cox

LEHNINGER

PRINCIPLES OF BIOCHEMISTRY

Sixth Edition

CHAPTER 15

Principles of Metabolic Regulation

Today, in this class...



Regulation of
glycolysis &
gluconeogenesis



Glycogen
breakdown &
synthesis

15.3 Regulation of Glycolysis and Gluconeogenesis

Glycolysis is under tight regulation

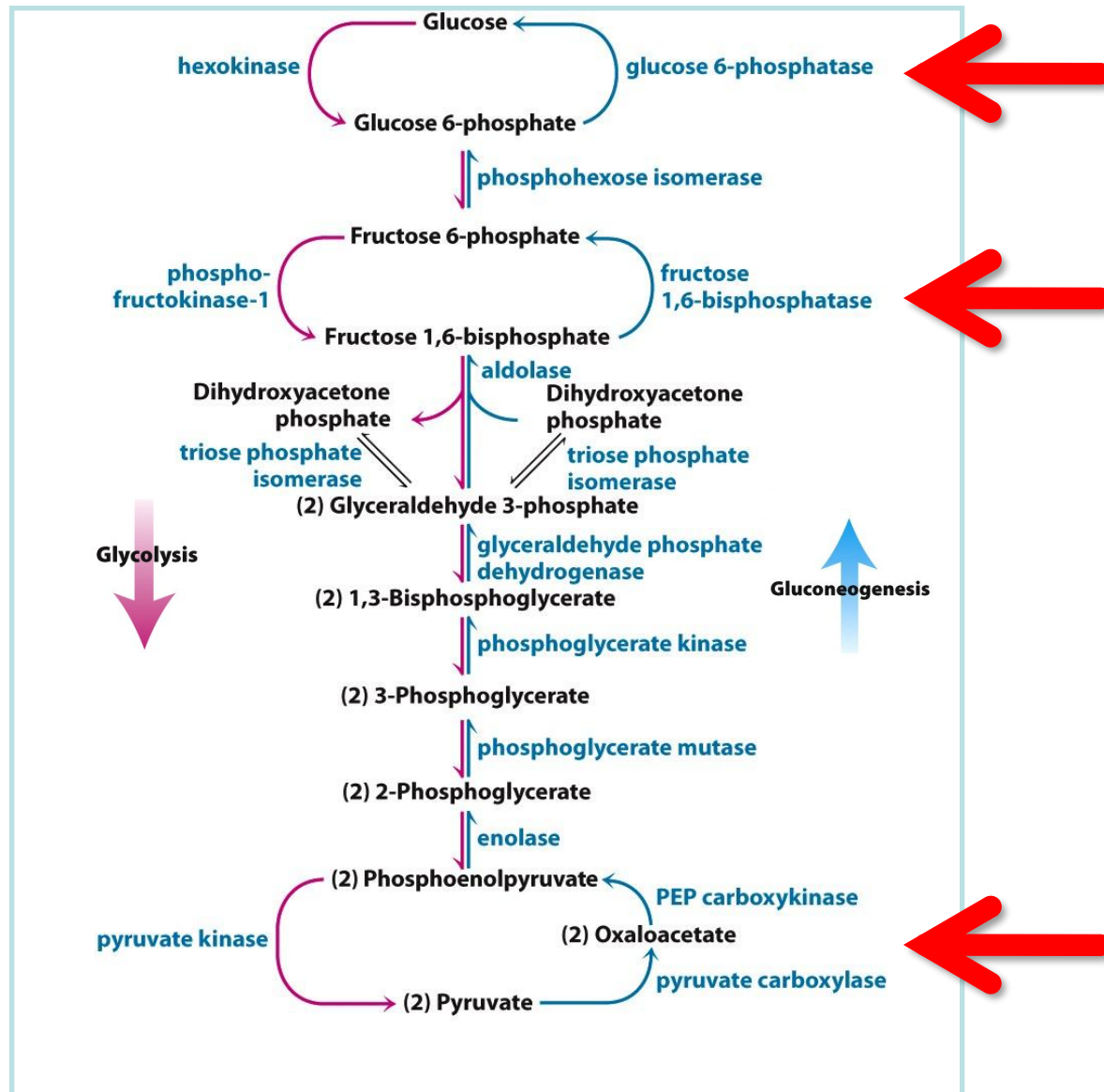
Regulation of glycolysis to maintain;

ATP levels, glycolytic intermediates for biosynthesis

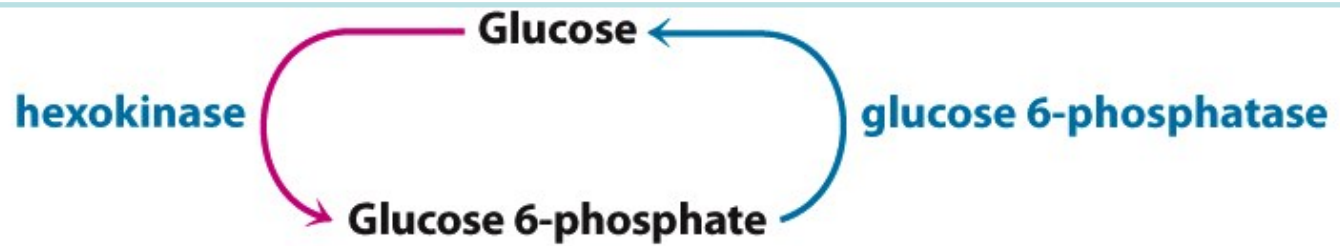
Regulation of glycolysis is achieved by;

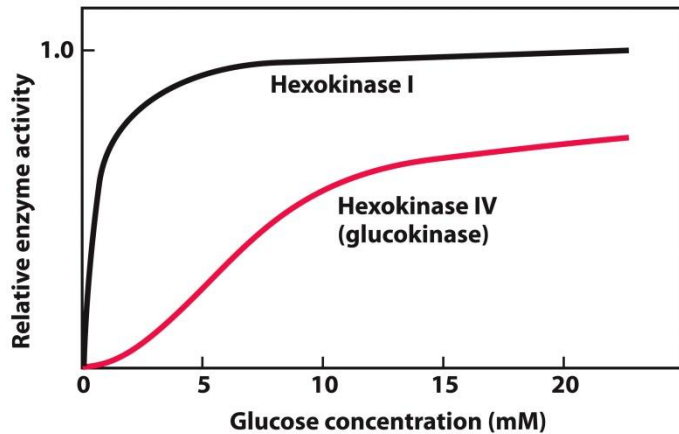
ATP consumption, NADH regeneration, allosteric regulation of glycolytic enzymes, and conc of metabolites

Glycolysis vs. Gluconeogenesis



Step 1





Hexokinase

Blood $[G] = 4\sim 5$ mM

Hexokinase I, II in muscle :

$K_m = 0.1$ mM

; **always saturated**

→ Always acting in maximal rate to produce G-6-P

→ Allosterically regulated by [G-6-P]



Hexokinase IV in liver :

$K_m = 10$ mM

; **half-saturated**

→ Hk IV is regulated by the level of blood glucose

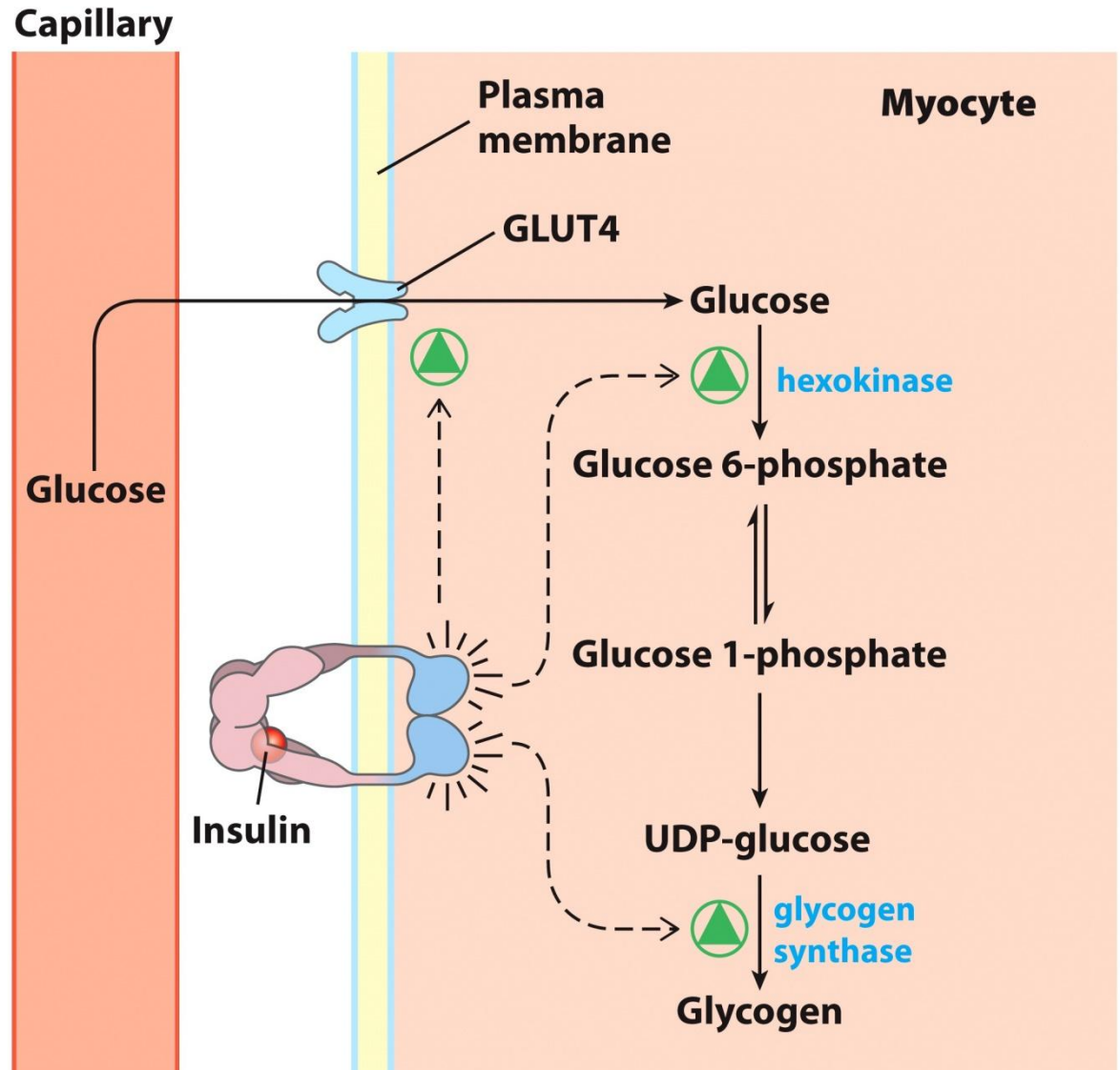


in muscle

If [G] up after meal,

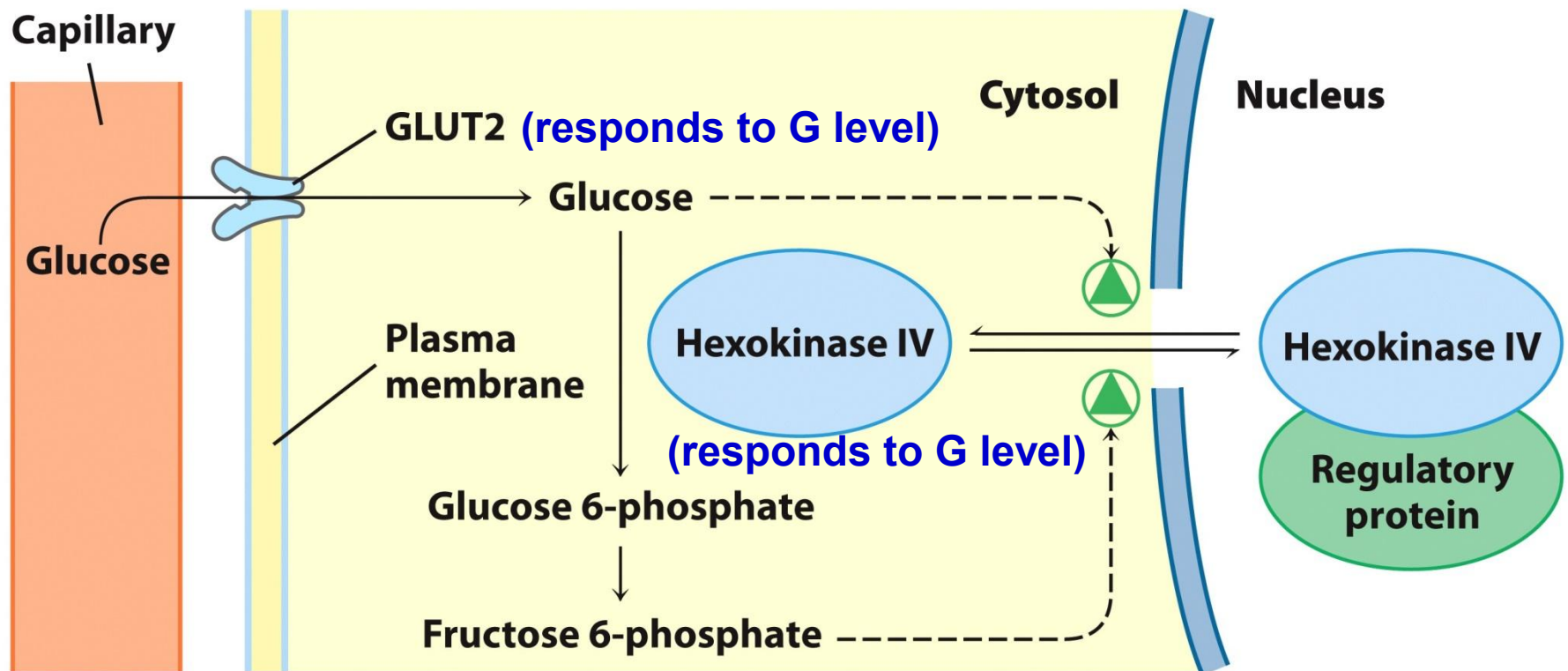
Glut4 respond to insulin
Glycolysis stimulated
Glycogen synthesis up

Hk II is regulated by
[G6P]



in liver

- Hexokinase IV is regulated by [G] and [F6P]
(Reversible binding of regulatory protein)
- Feeding : high **G** → active Hk IV /
- Fasting : **F6P** binds → inactive Hk IV



The initial rate of glucose entry into an erythrocyte, V_0 , depends on the initial concentration of glucose on the outside, $[S]_{\text{out}}$.

Blood glucose ($[S]_{\text{out}}$) : $\sim 5\text{mM}$

K_t of GLUT1 (erythrocyte) : 1.5mM

GLUT1 is nearly saturated with substrate, high transport rates

GLUT2 (liver) : $K_t = \sim 66\text{mM}$

GLUT2 respond to increased levels of intracellular glucose, produced by glycogen breakdown

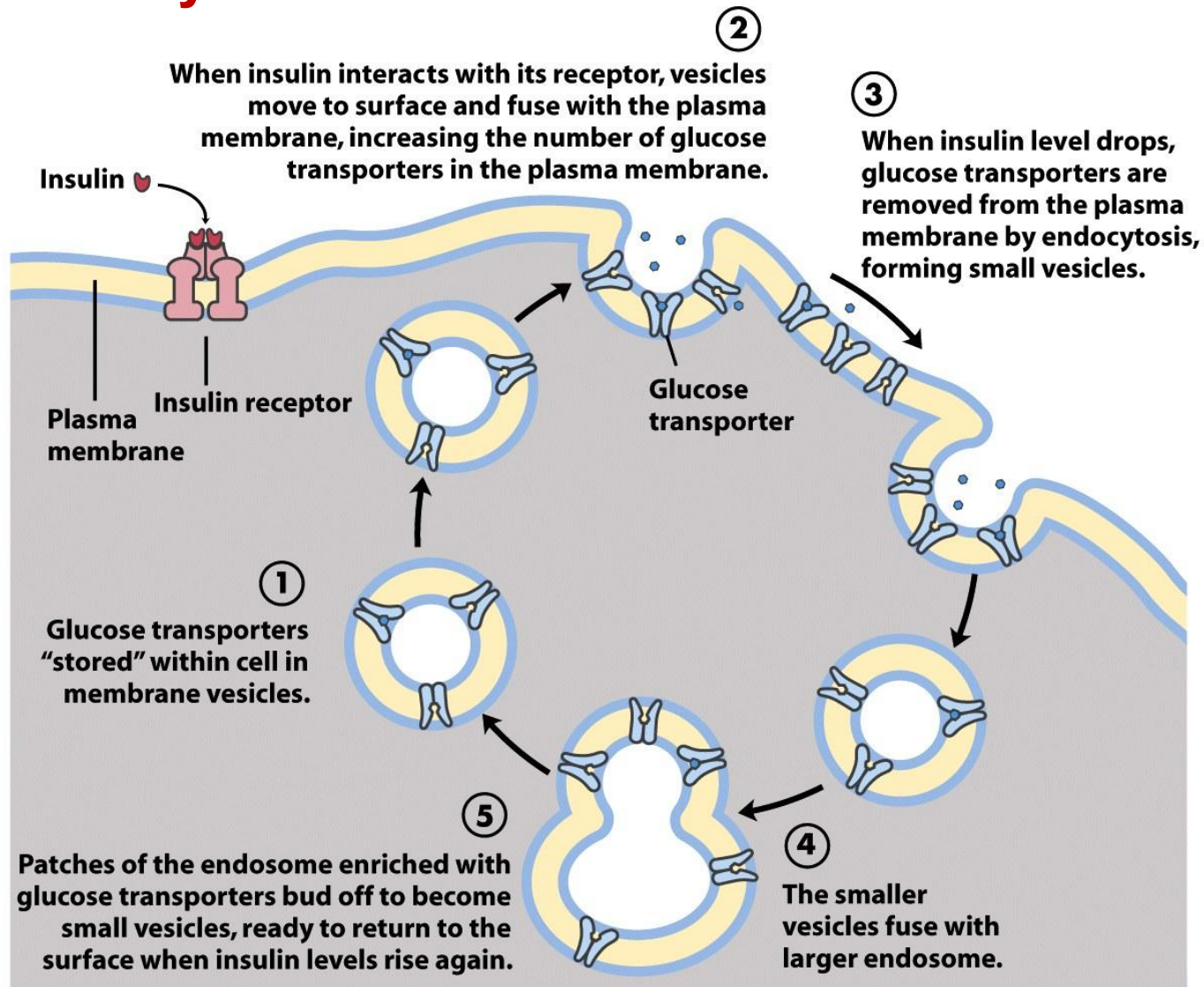
GLUT3 (brain) : high affinity type

allowing it to transport even in times of low glucose conc.

GLUT4 (muscle & adipocyte) : $K_t = 5\text{mM}$

GLUT4 respond to insulin

Transport of glucose into a myocyte by GLUT4 is regulated by insulin



Box 11-2 figure 1

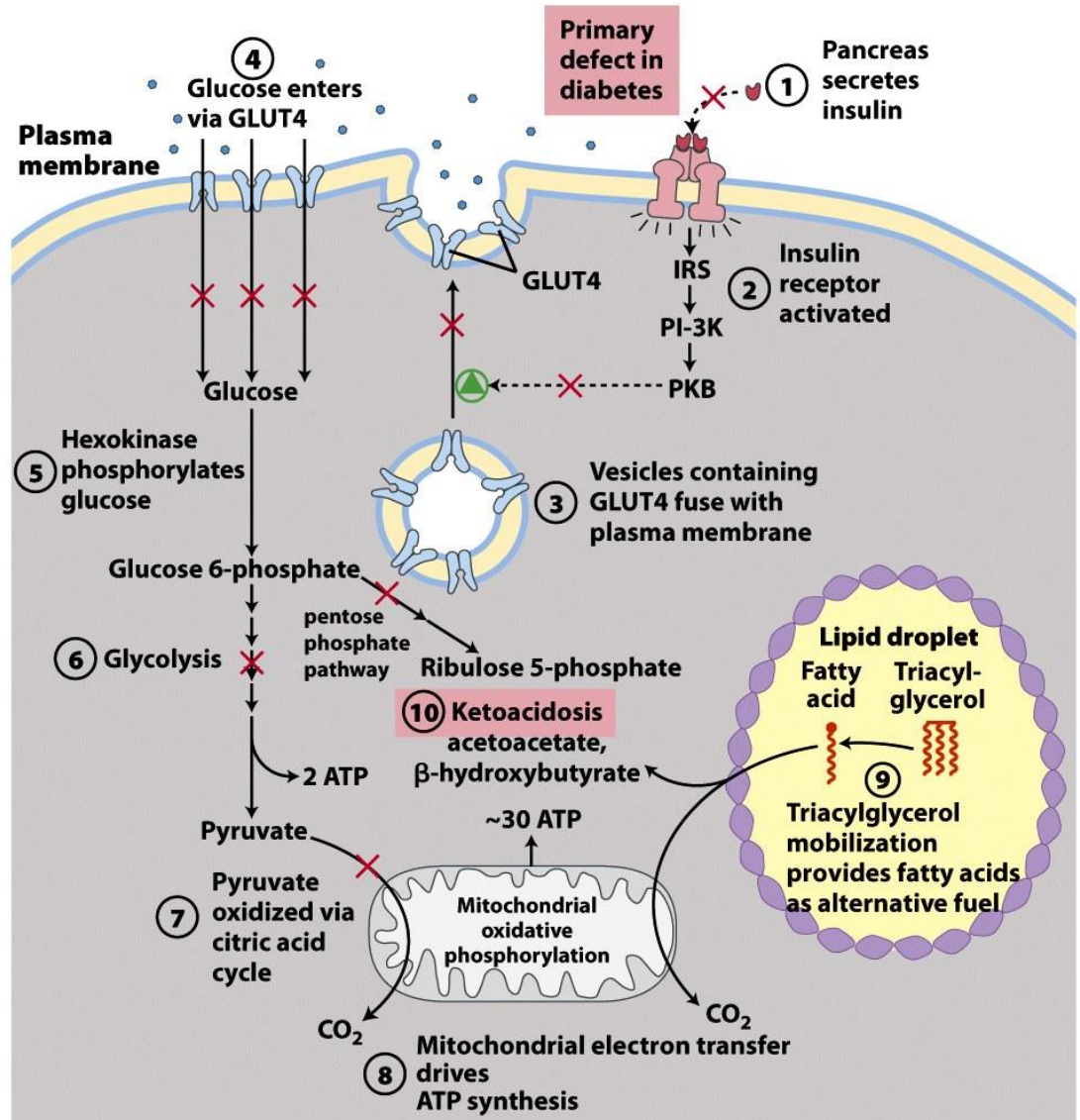
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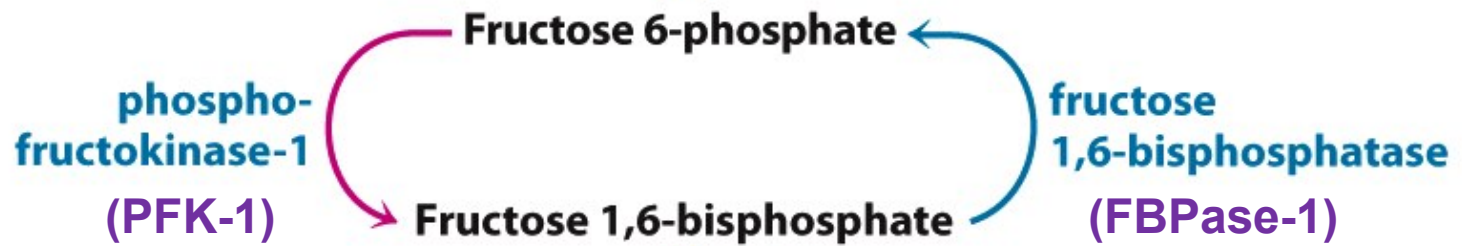
Effect of type 1 diabetes on carbohydrate and fat metabolism in an adipocyte

In muscle and adipose tissues, **GLUT4**, sequestered in vesicles, moves into the plasma membrane in response to an **insulin** signal

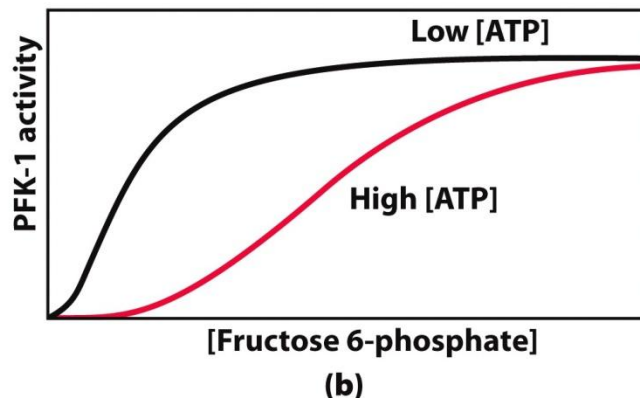
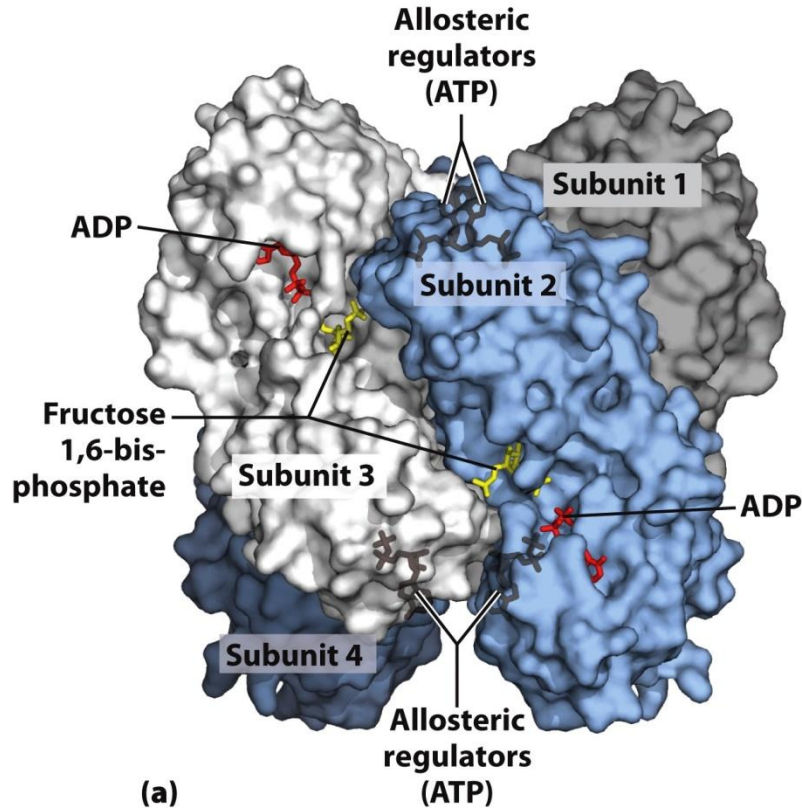
In type 1 diabetes, muscle and fat tissue use fatty acids as an alternative fuel
→ Accumulation of “ketone bodies”
→ **ketoacidosis**



Step 3

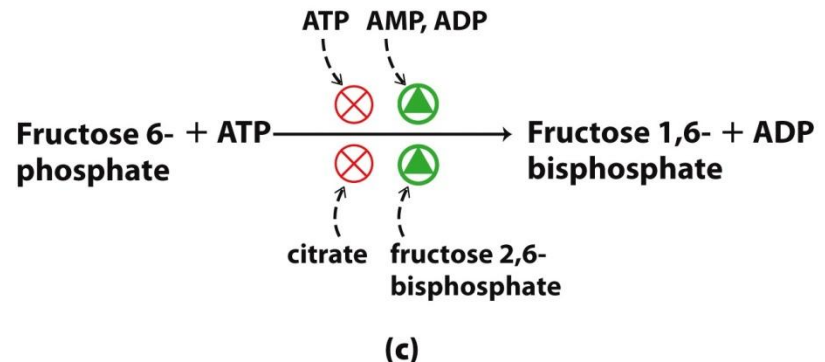


PFK-1 and its regulation

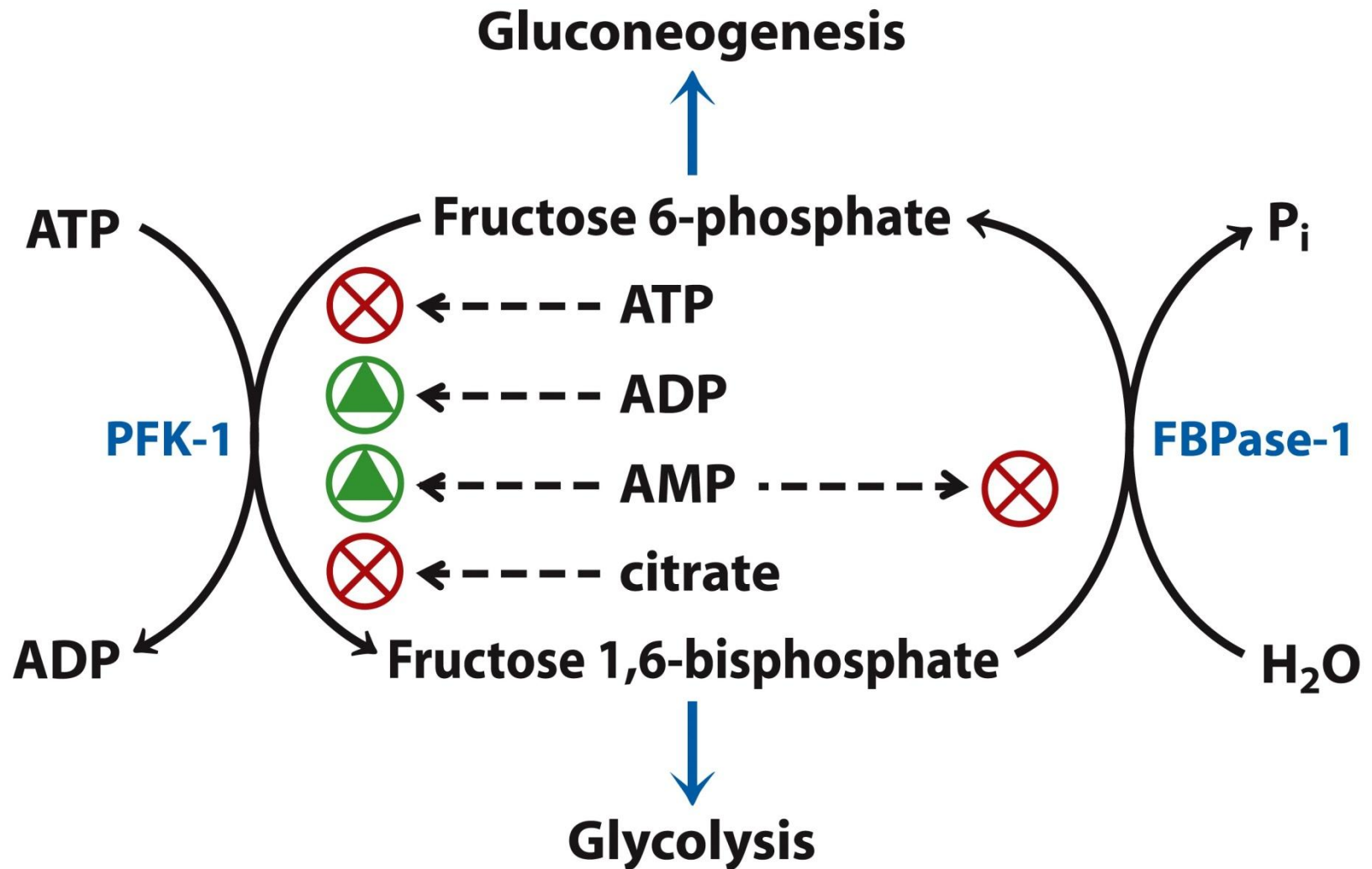


ATP is a substrate of PFK-1
ATP is a product of glycolysis

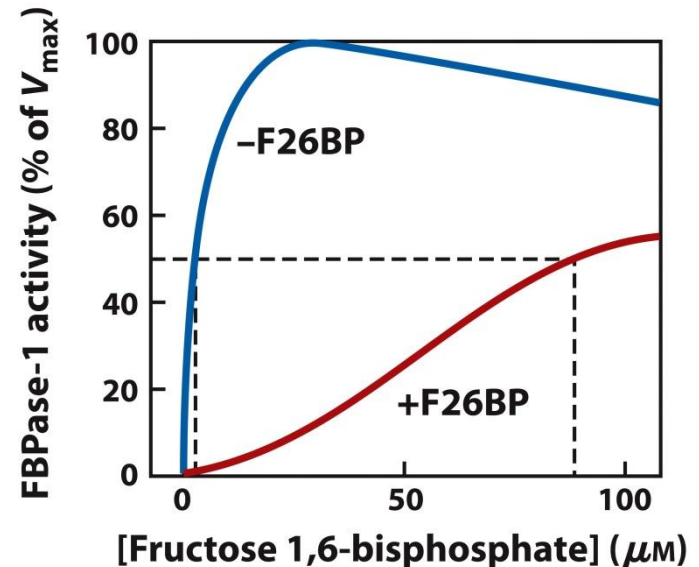
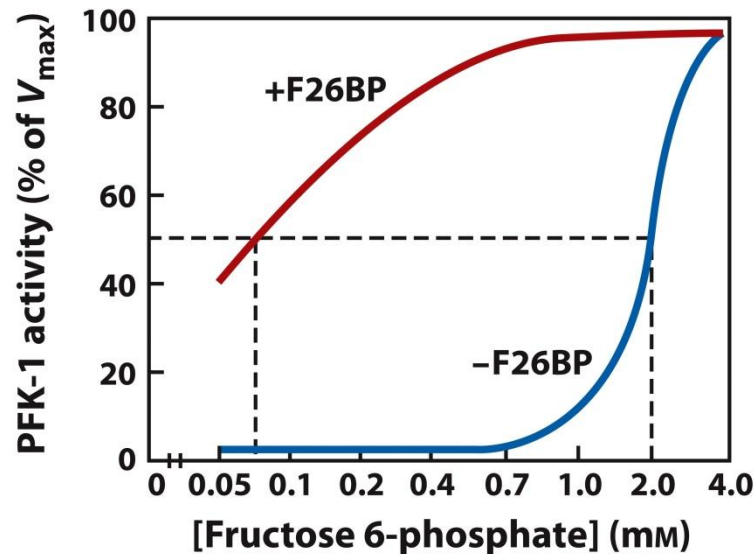
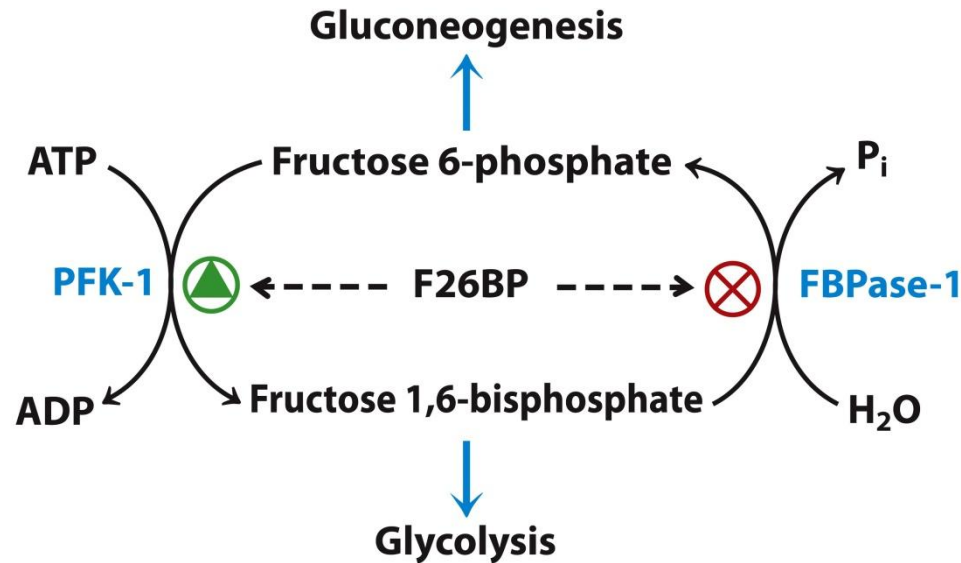
$\uparrow[\text{ATP}]$: binds allosteric site
 $\rightarrow$ Inhibit PFK-1



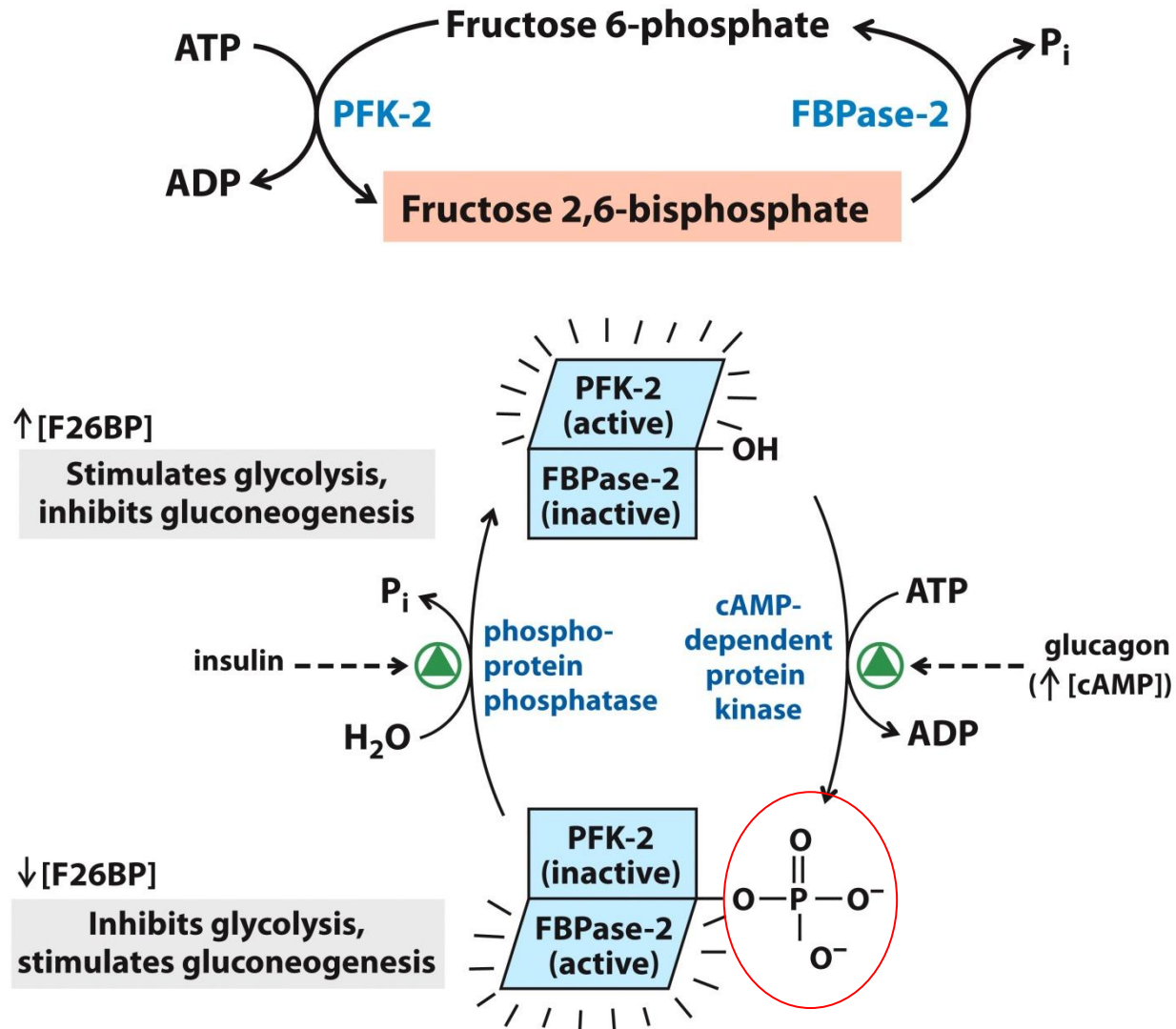
PFK-1 and FBPase-1 is reciprocally regulated



F26BP activate the PFK-1 and inhibit FBPase-1



Regulation of F26BP level by hormones



Xylulose 5-P is a key regulator of F26BP



High carbohydrate diet



↑ Pentose P pathway



↑ Xylulose 5-P



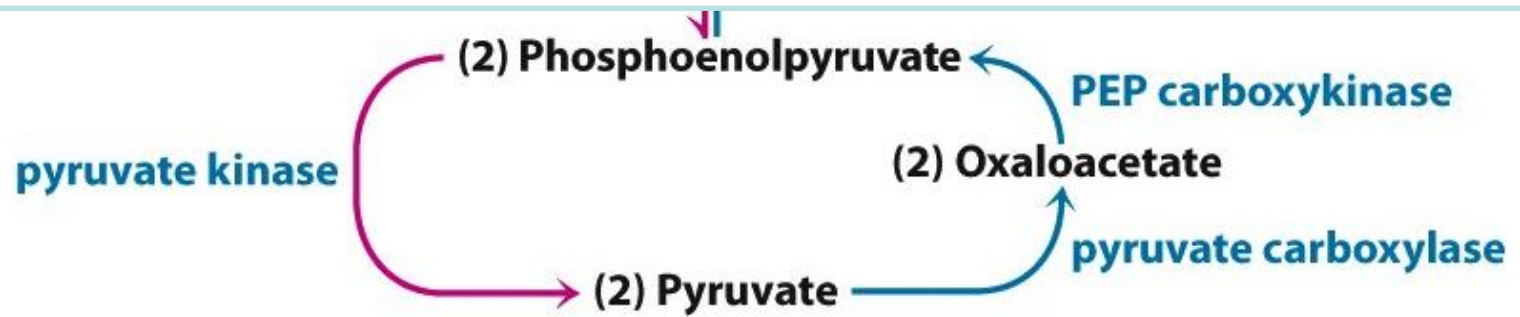
PP2A activated → deP of PFK-2/FBPase-2 → ↑ F26BP



↑ Glycolysis, ↑ FA synthesis

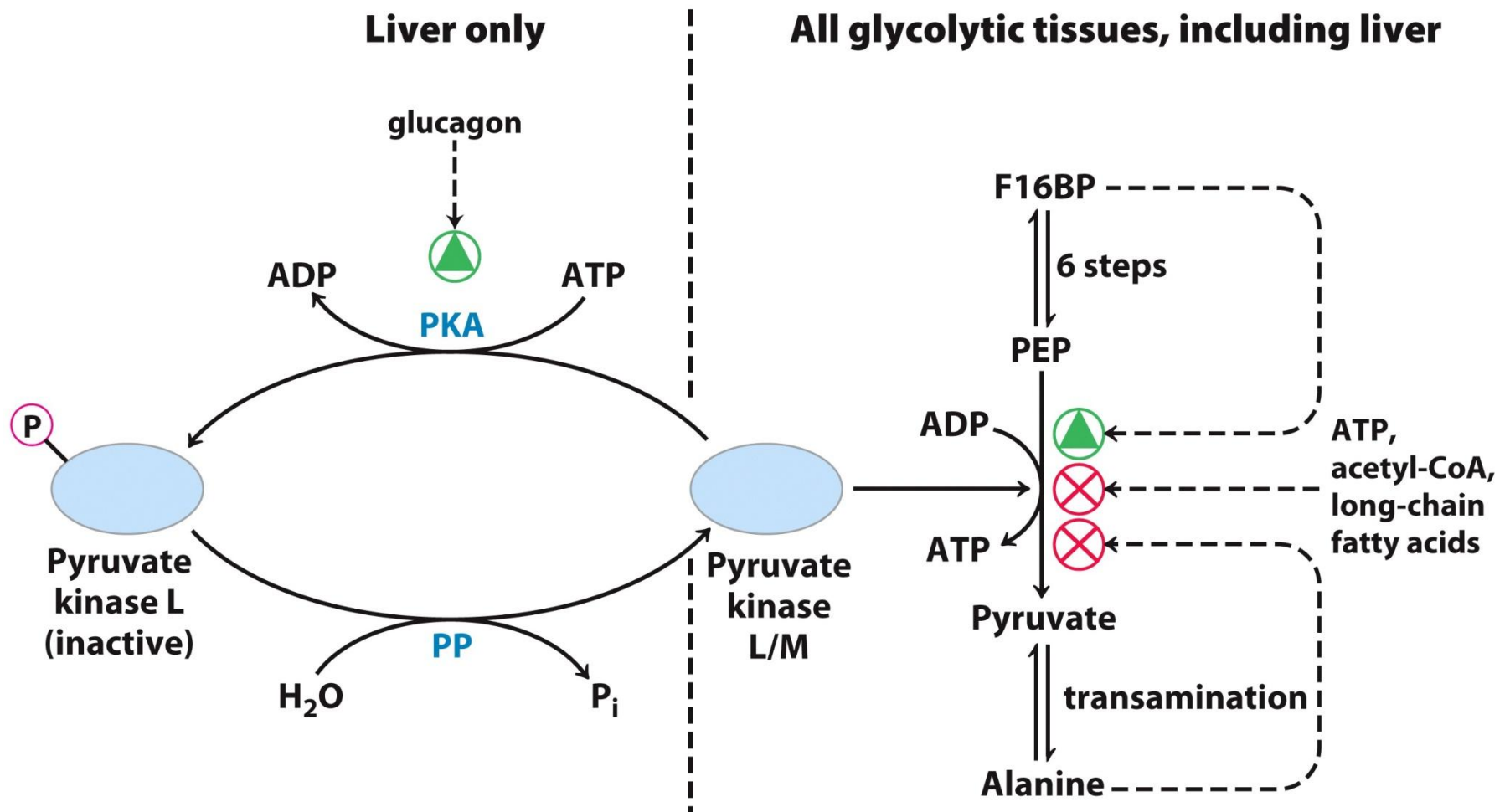


Step 10

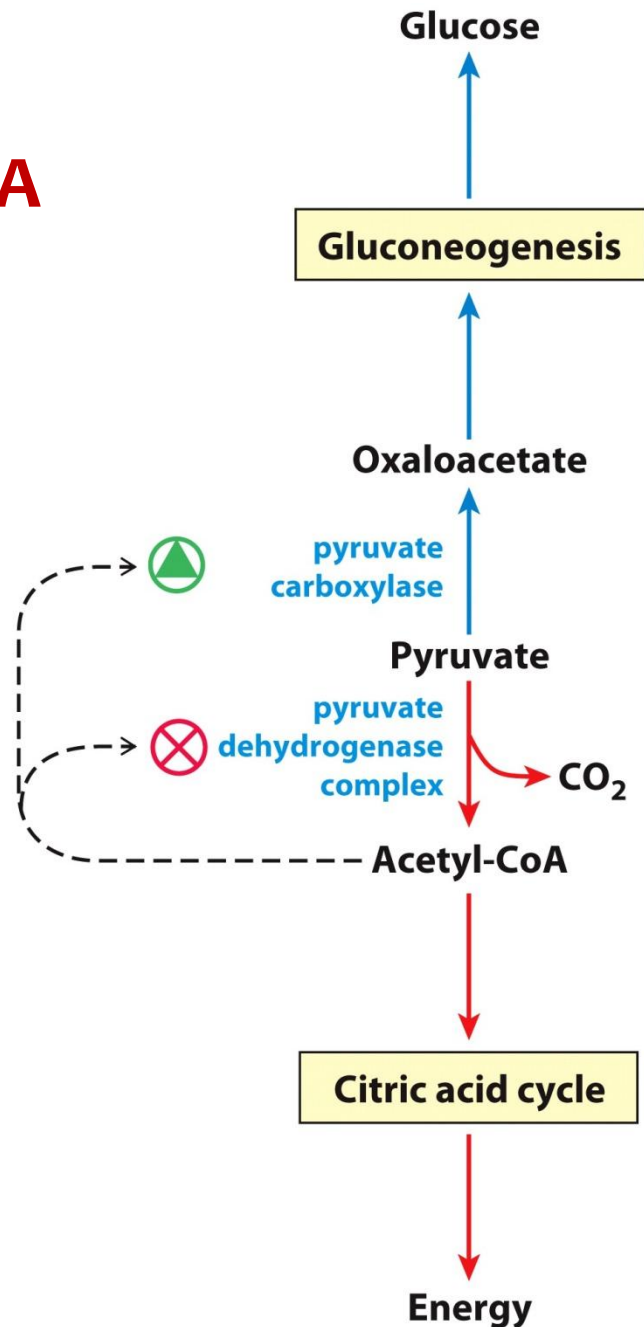


Regulation of pyruvate kinase

In liver, low [G] $\rightarrow$ PK inactivated by PKA $\rightarrow$ glucose usage $\downarrow$



Regulation of pyruvate metabolism by **Acetyl-CoA**



Transcriptional regulation of Hexokinase and G-6-Phosphatase

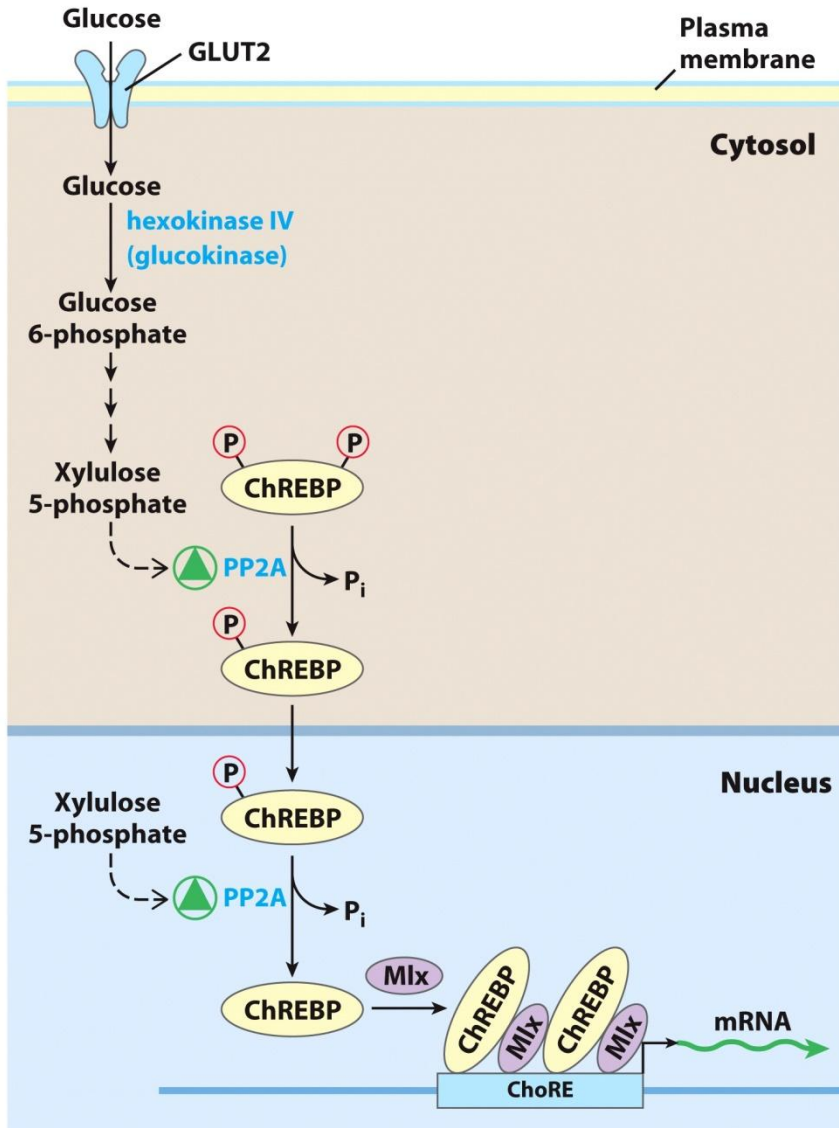
For greater energy production (low ATP, vigorous exercise)
Or for greater glucose consumption (high blood G)

→ Increased transcription of hexokinase IV

For increased production of glucose (low blood G)

→ Increased transcription of G-6-phosphatase

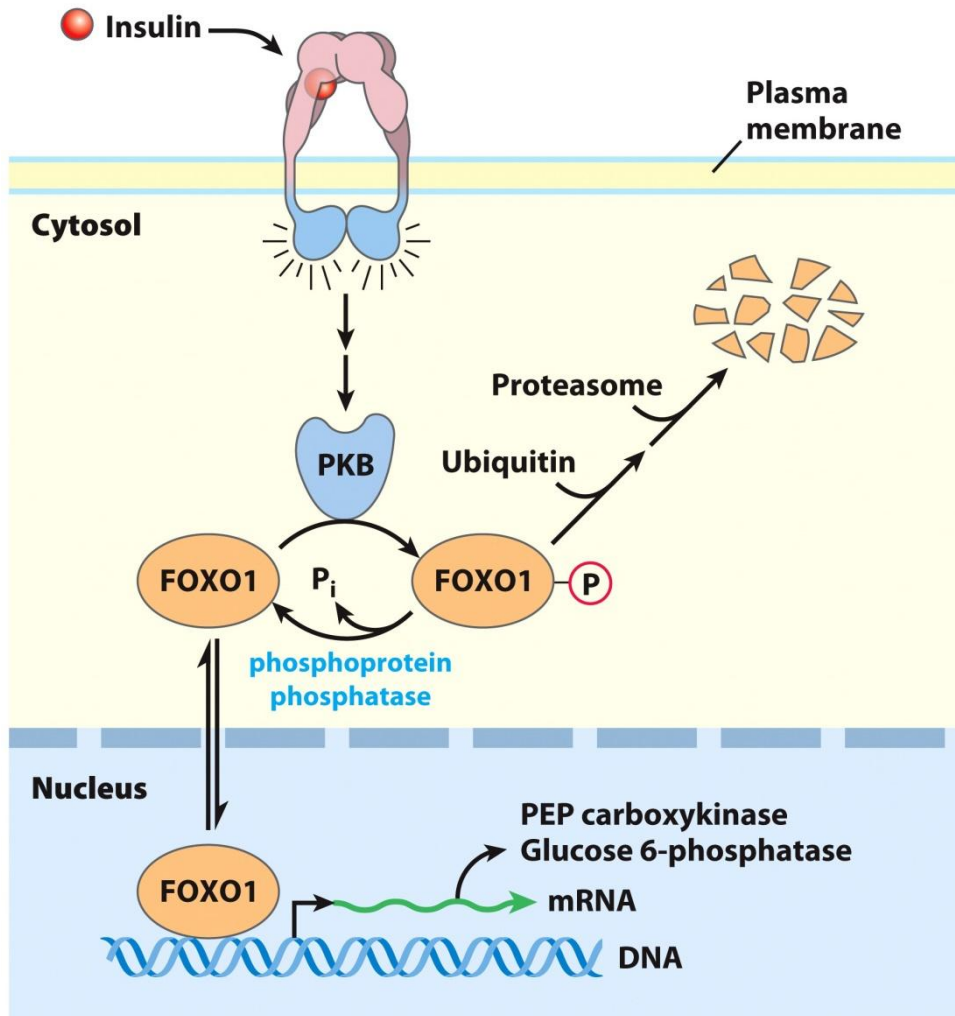
Transcriptional regulation of ChREBP (glycolysis ↑)



ChREBP (carbohydrate response element binding protein) &

: ↑ **Enzymes for glycolysis**

Transcriptional regulation of FOXO1 (gluconeogenesis ↑)

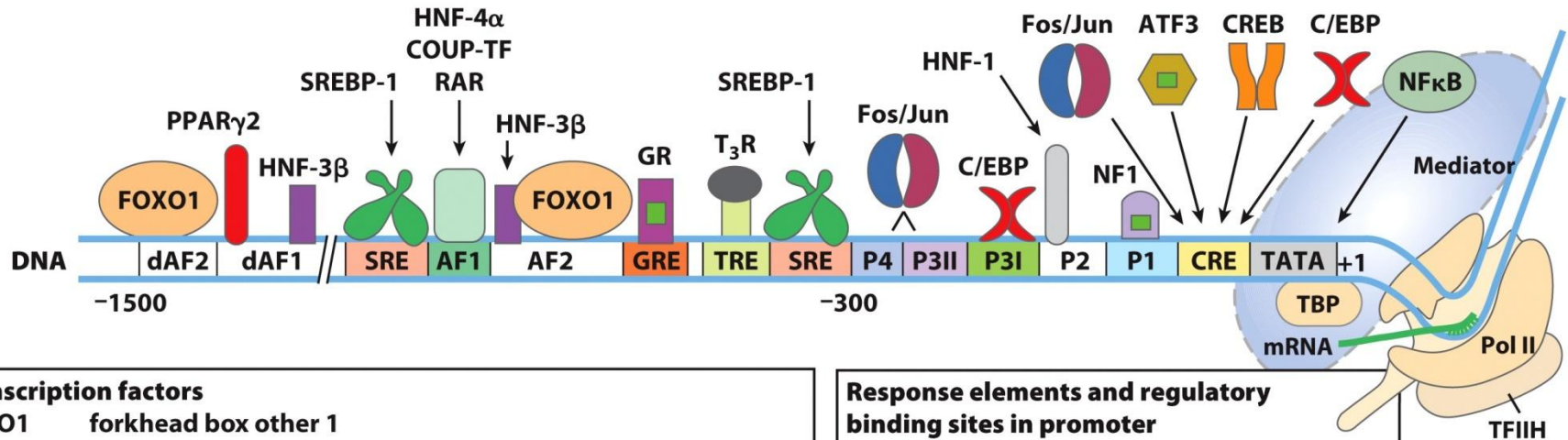


→ Resulting in increase of glycolysis

FOXO1 (forkhead box other 1)

→ Stimulate gluconeogenesis

Transcriptional regulation : complexity of regulation !!



Transcription factors

FOXO1	forkhead box other 1
PPAR γ 2	peroxisome proliferator-activated receptor γ 2
HNF-3 β	hepatic nuclear factor-3 β
SREBP-1	sterol regulatory element binding protein-1
HNF-4 α	hepatic nuclear factor-4 α
COUP-TF	chicken ovalbumin upstream promoter-transcription factor
RAR	retinoic acid receptor
GR	glucocorticoid receptor
T ₃ R	thyroid hormone receptor
C/EBP	CAAT/enhance binding protein
HNF-1	hepatic nuclear factor-1
NF1	nuclear factor 1
ATF3	activating transcription factor 3
CREB	cAMP response element binding protein
NF κ B	nuclear factor κ B
TBP	TATA-box binding protein
TFIIH	transcription factor IIH

Response elements and regulatory binding sites in promoter

dAF2	distal accessory factor 2
dAF1	distal accessory factor 1
SRE	sterol regulatory element
AF1	accessory factor 1
AF2	accessory factor 2
GRE	glucocorticoid regulatory element
TRE	thyroid hormone regulatory element
CRE	cAMP regulatory element

The PEP carboxykinase promoter region

Figure 15-25

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Summary

1. The points of regulation in glycolysis & gluconeogenesis
 - 3 reactions
2. Hexokinase isozymes of muscle and liver are affected differently by G6P
3. Hexokinase IV and G6Pase are transcriptionally regulated
4. PFK1 and FBPase1 are reciprocally regulated
5. F26BP is an allosteric regulator of PFK1 and FBPase1
6. Xu5P is a key regulator of PP2A
7. Pyruvate kinase is allosterically regulated by ATP
8. Regulation of pyruvate to PEP for gluconeogenesis
9. Transcriptional regulation of glycolysis and gluconeogenesis

Today, in this class...



Regulation of
glycolysis &
gluconeogenesis

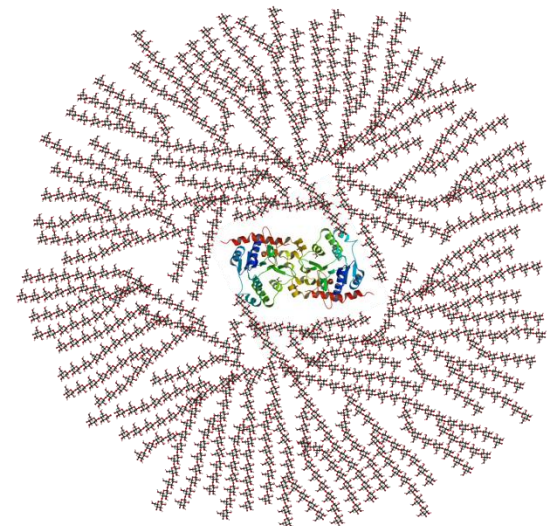
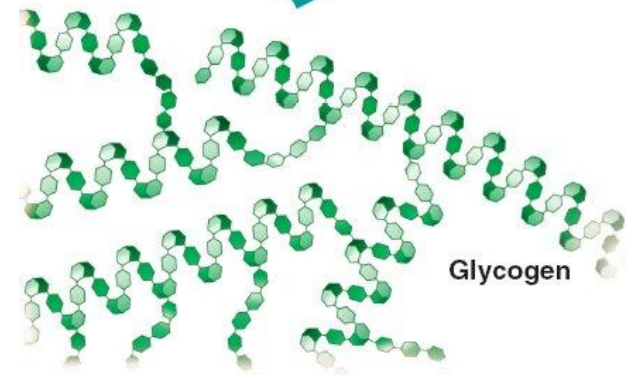
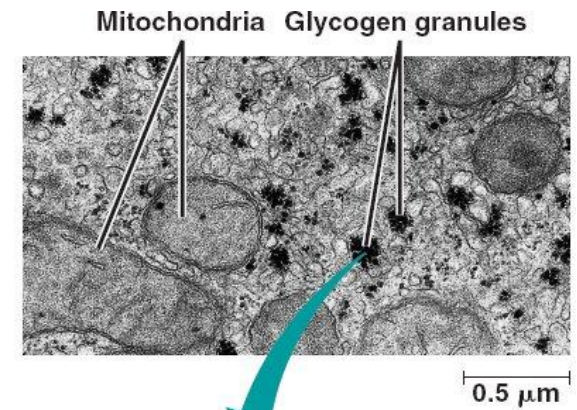
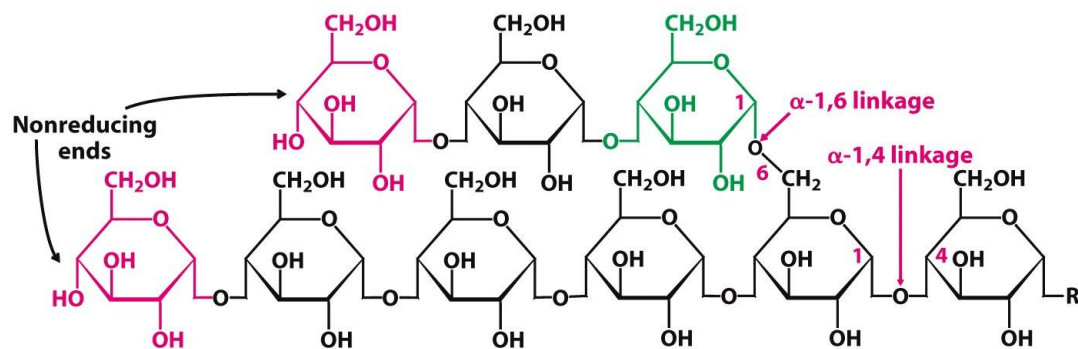


Glycogen
breakdown &
synthesis

15.4 The metabolism of glycogen in animals

The diagram illustrates the metabolic pathways of carbohydrates in the human body, centered around a silhouette of a person. The pathways are color-coded and labeled as follows:

- Intake:** Represented by a red arrow pointing into the body from the top left.
- Starch:** Shown as a branched polymer of glucose units.
- Glycogen:** Shown as a highly branched, star-shaped polymer of glucose units.
- Storage:** Indicated by a green arrow pointing from the liver area to the muscle area, labeled "Storage: Glycogen".
- Distribution and utilization:** Indicated by a blue arrow pointing from the muscle area back to the liver area, labeled "Distribution and utilization: Free glucose".
- Disaccharides:** Shown as two glucose units linked together.
- Monosaccharides (glucose, fructose, galactose):** Shown as single glucose units.



Glycogen granules in a hepatocyte

Liver glycogen serves as a reservoir of glucose for other tissues
→ especially important for brain; can be depleted in 12~24 hrs

As glucose concentration: $\sim 0.4\text{M} \rightarrow 0.01 \mu\text{M}$ as glycogen

β -particle: 21 nm, $\sim 55,000$ glucose, 2,000 nonreducing ends
 α -rosette: 20~40 β -particles

Muscle glycogen
: for vigorous activity, can
be exhausted in < 1 hr

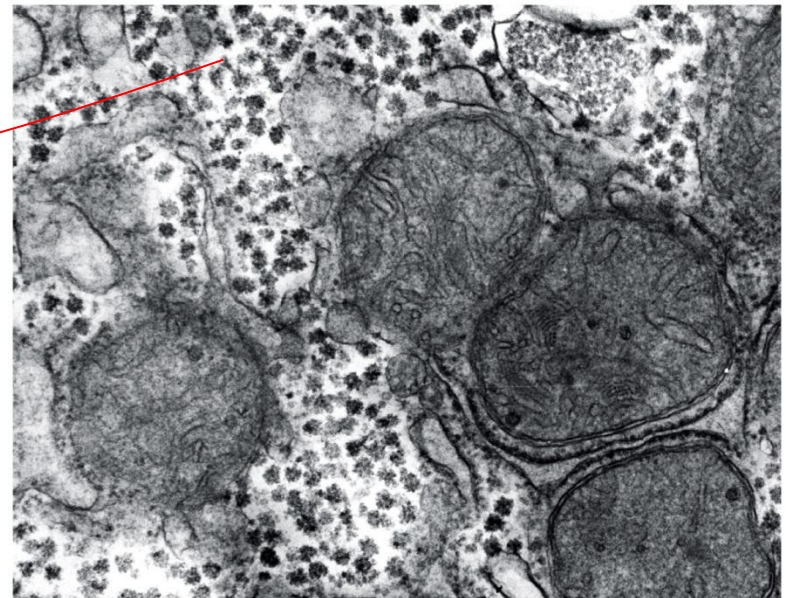
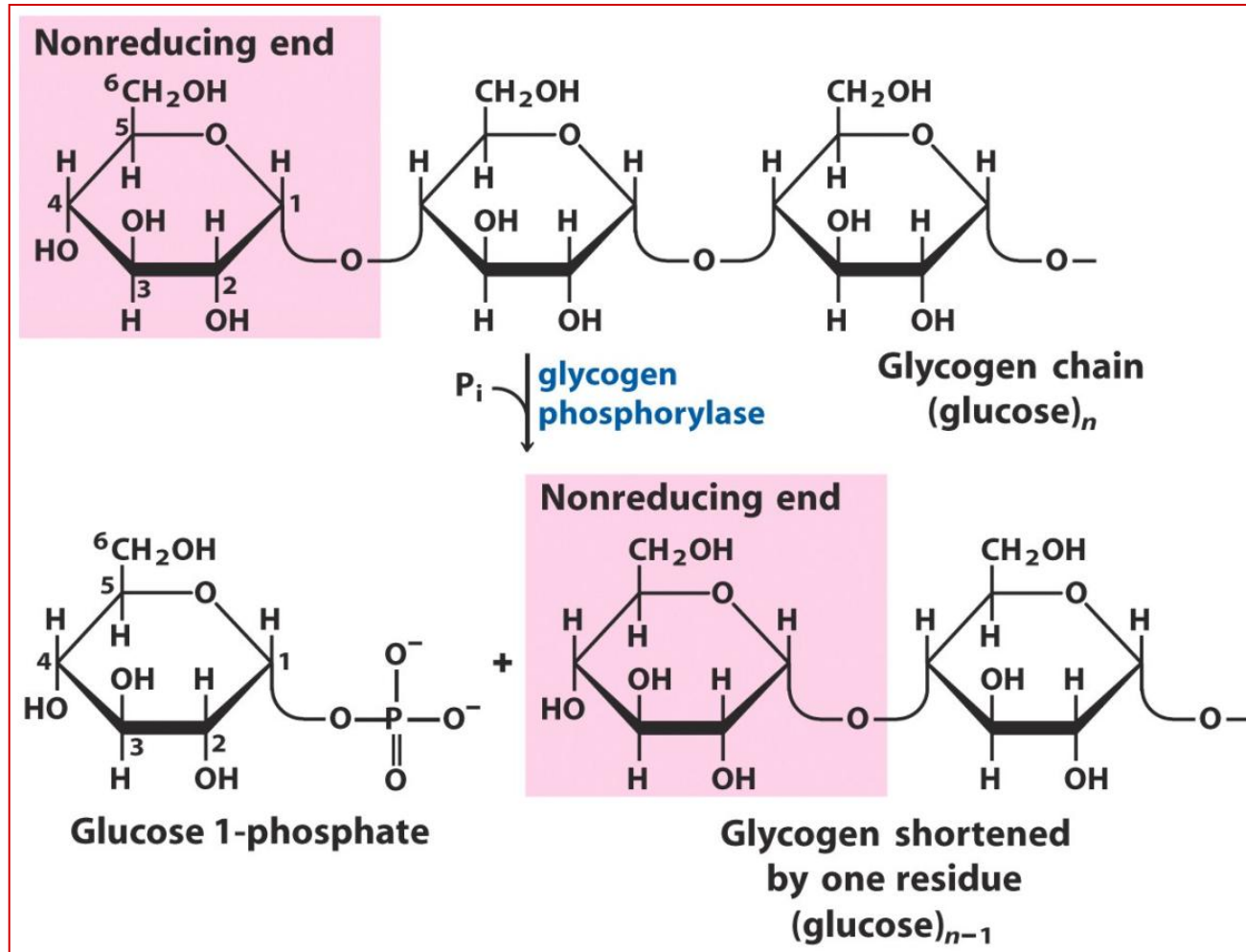


Figure 15-24
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Glycogenolysis (1)

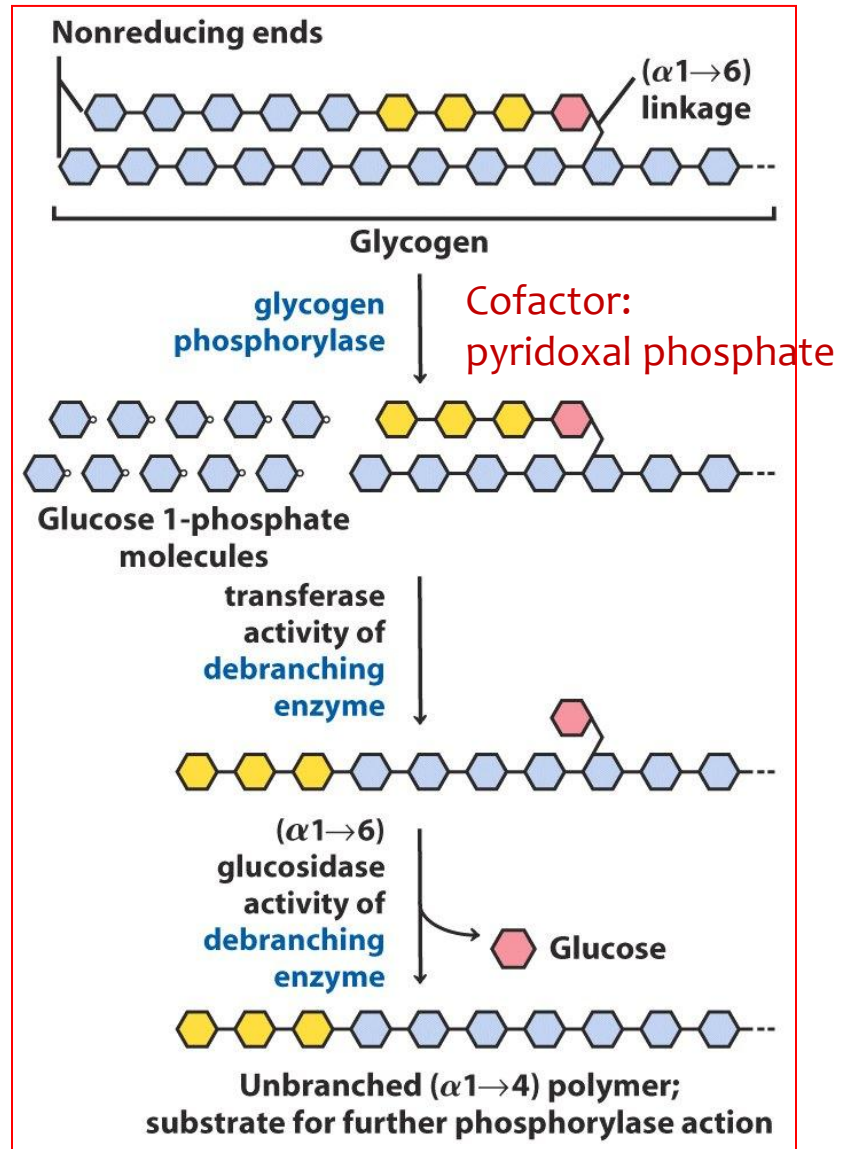
Glycogen phosphorylase breaks $\alpha 1,4$ glycosidic linkage at terminal non-reducing end: produce $(G)_{n-1}$ and G-1-P



Glycogenolysis (2)

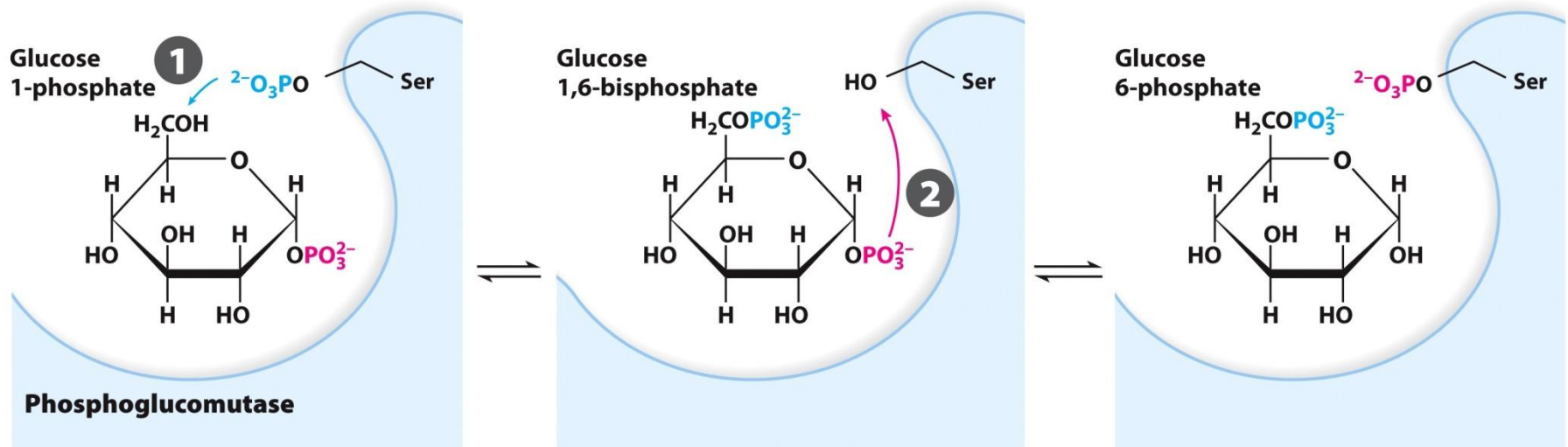
Debranching enzyme

1. **Transferase activity**
2. **Glucosidase activity**



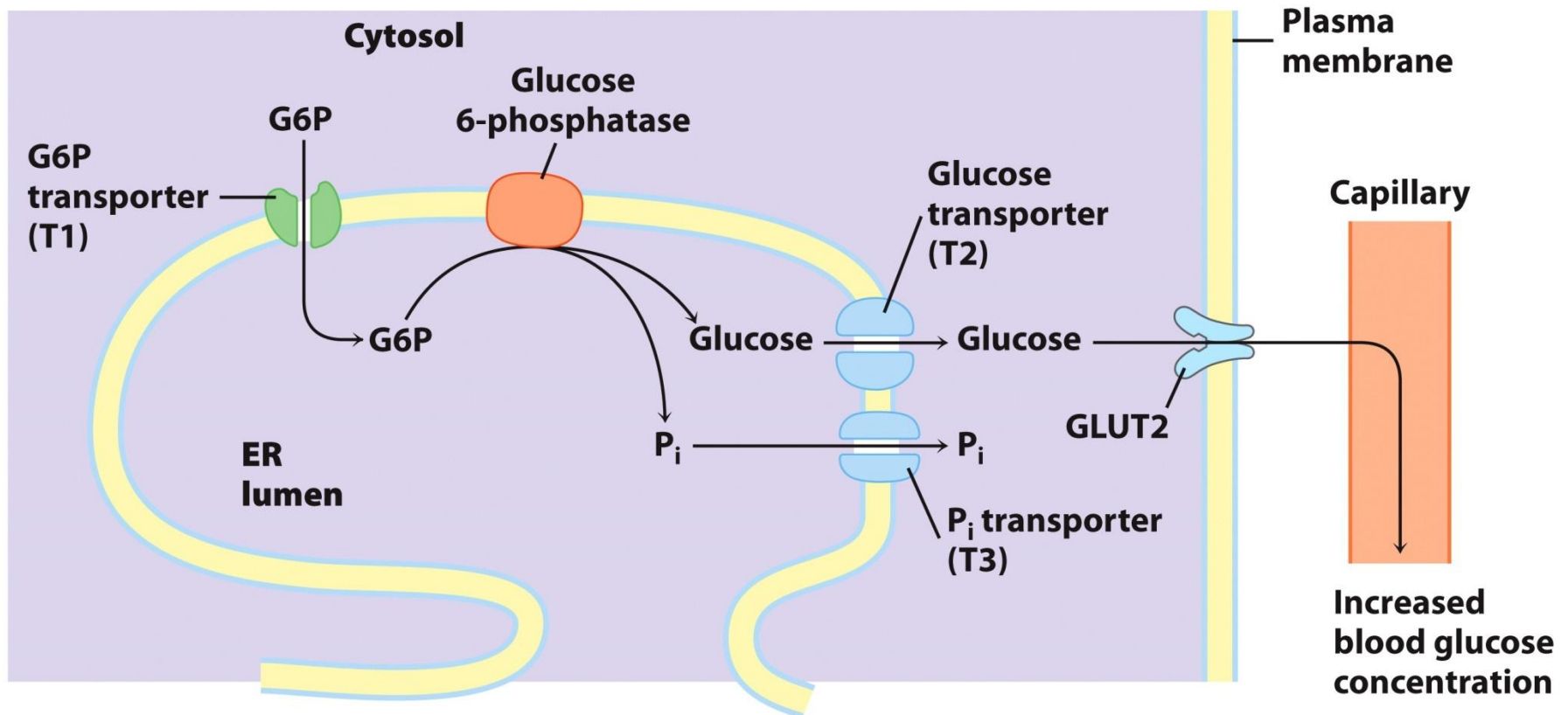
Glycogenolysis (3)

Phosphoglucomutase convert G1P to G6P



Glycogenolysis (4)

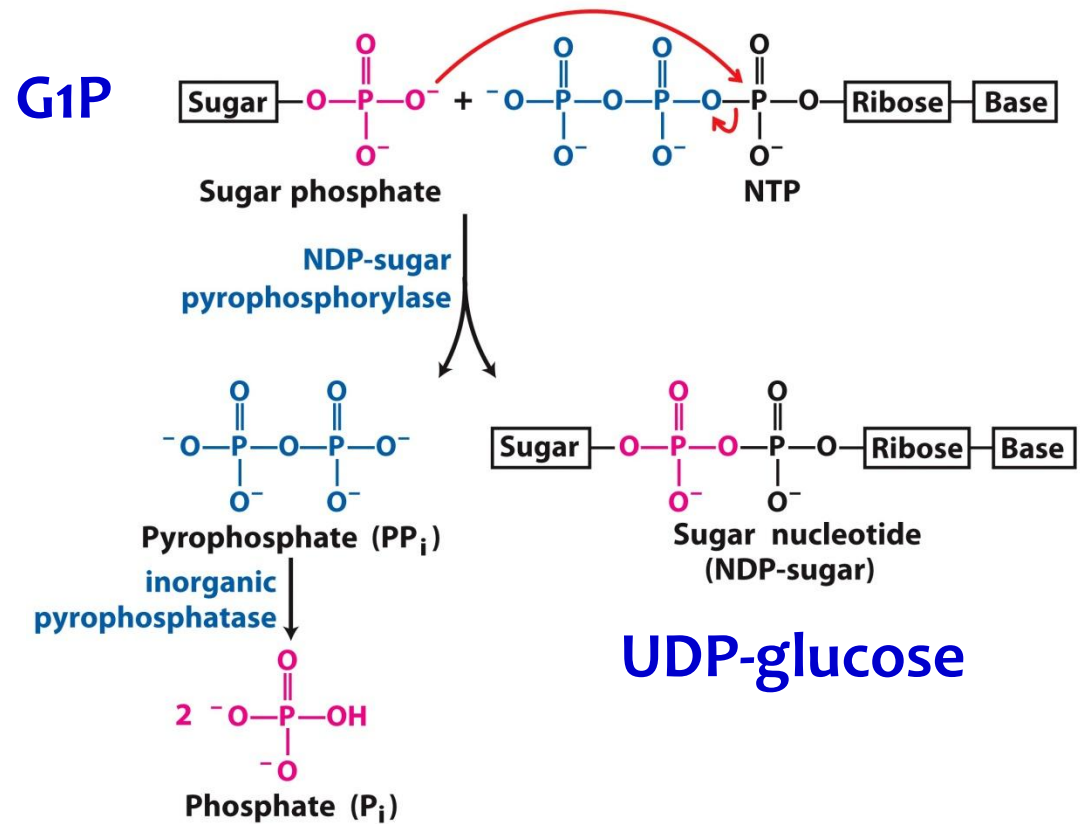
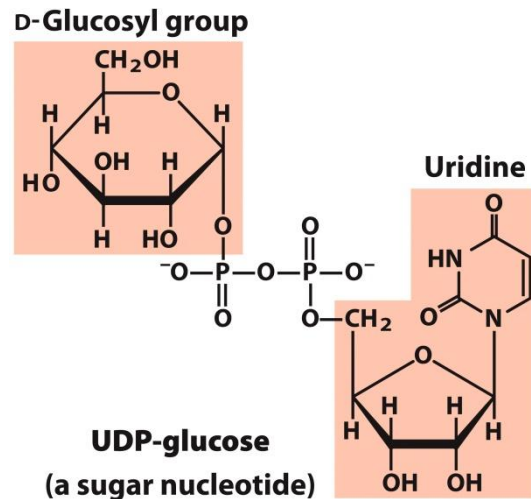
G6P is converted to glucose in ER
glucose is transported and released to blood



Defect in G-6-Pase : type Ia glycogen storage disease ('22 약시)

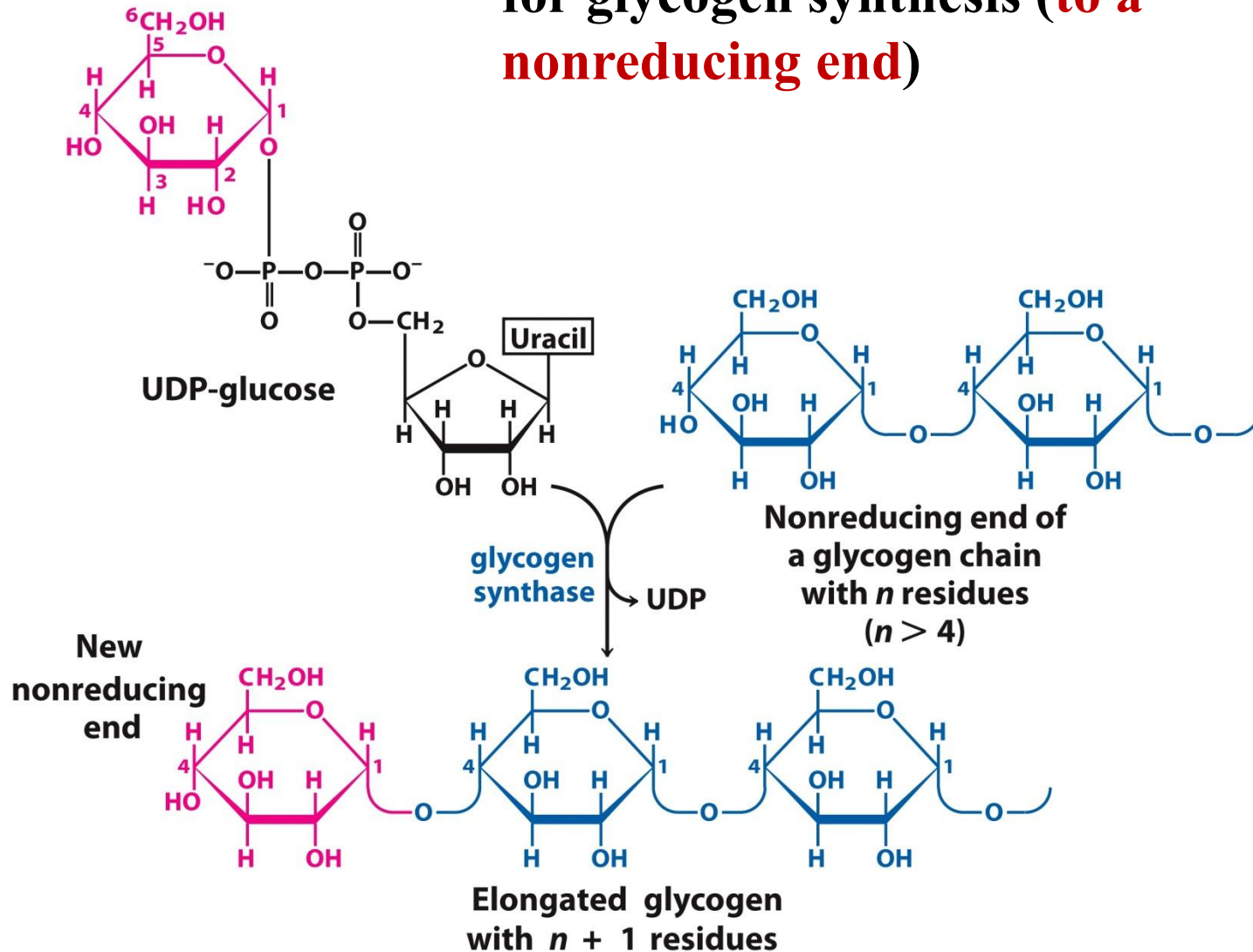
Glycogenesis (1)

Formation of UDP-glucose (uridine diphosphate glucose)



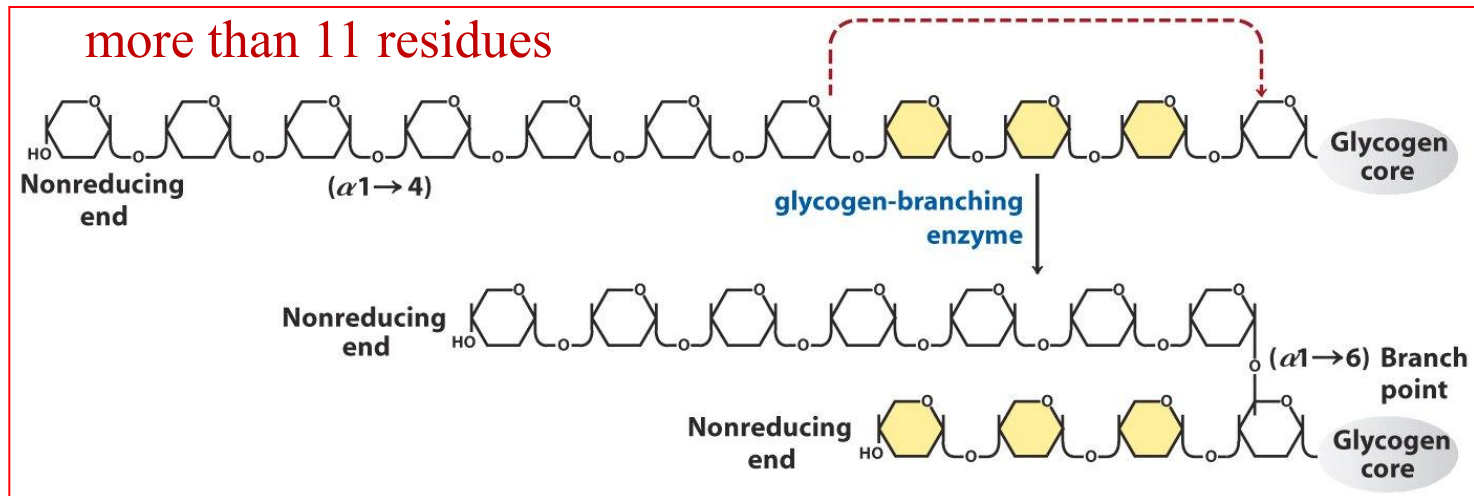
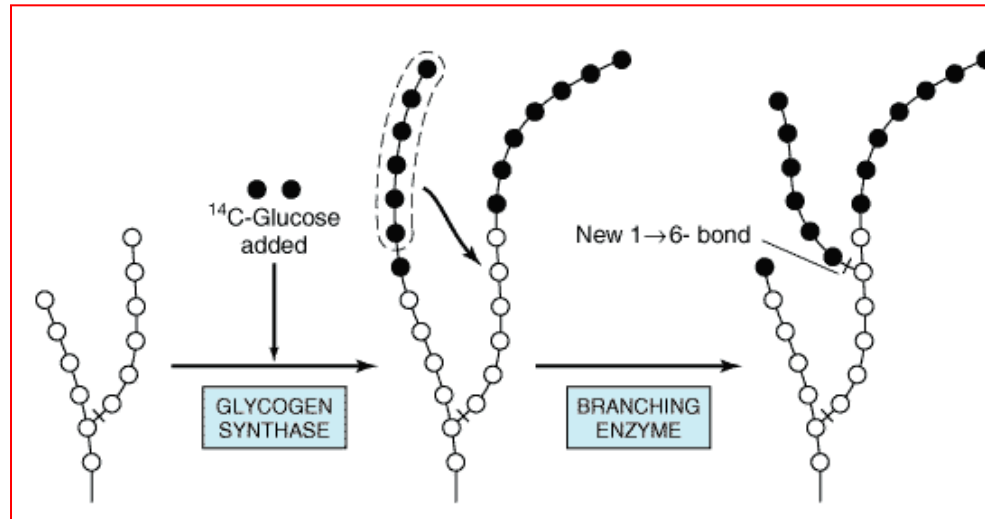
Glycogenesis (2)

UDP glucose donates glucose
for glycogen synthesis (**to a
nonreducing end**)



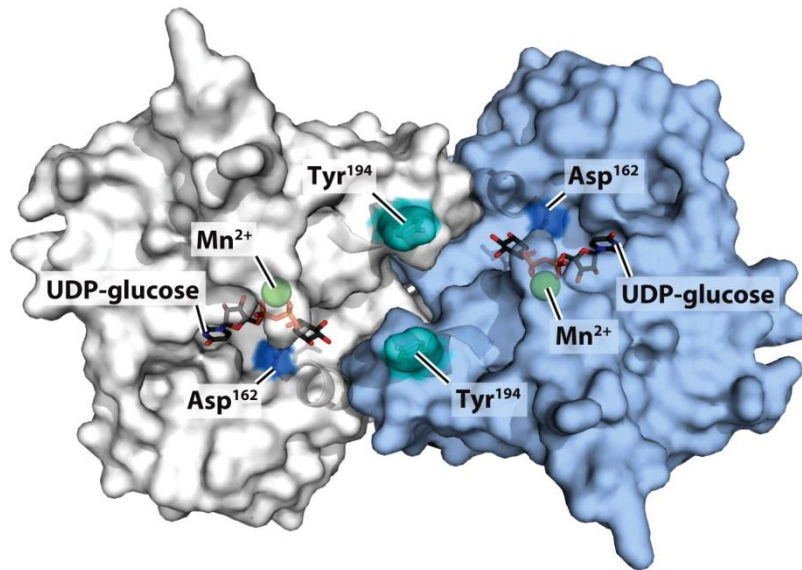
Glycogenesis (3)

Branching the glycogen to make ($\alpha 1 \rightarrow 6$) bonds

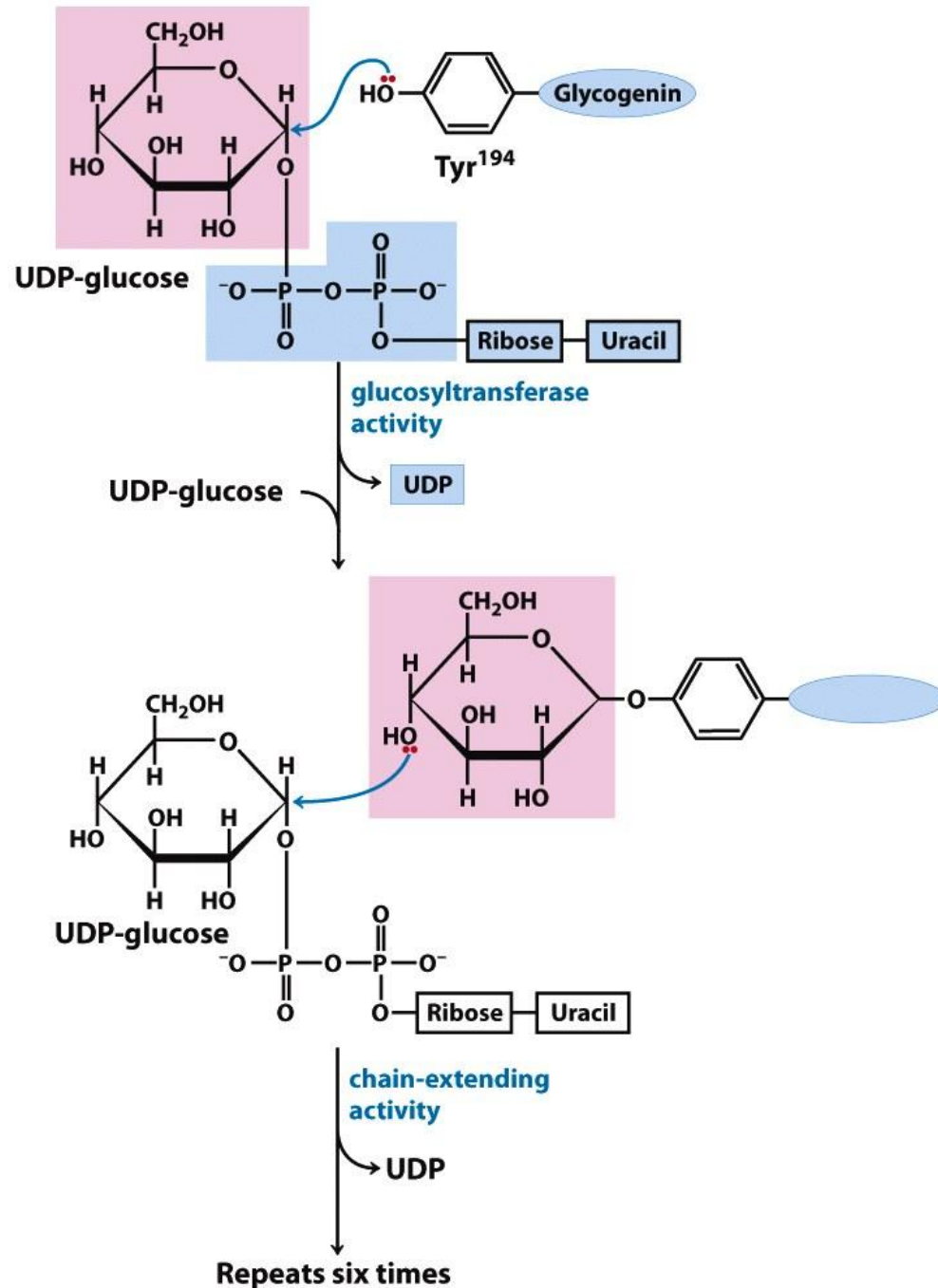


Glycogenesis (4)

glycogenin primes the initial sugar (up to 8 residues)

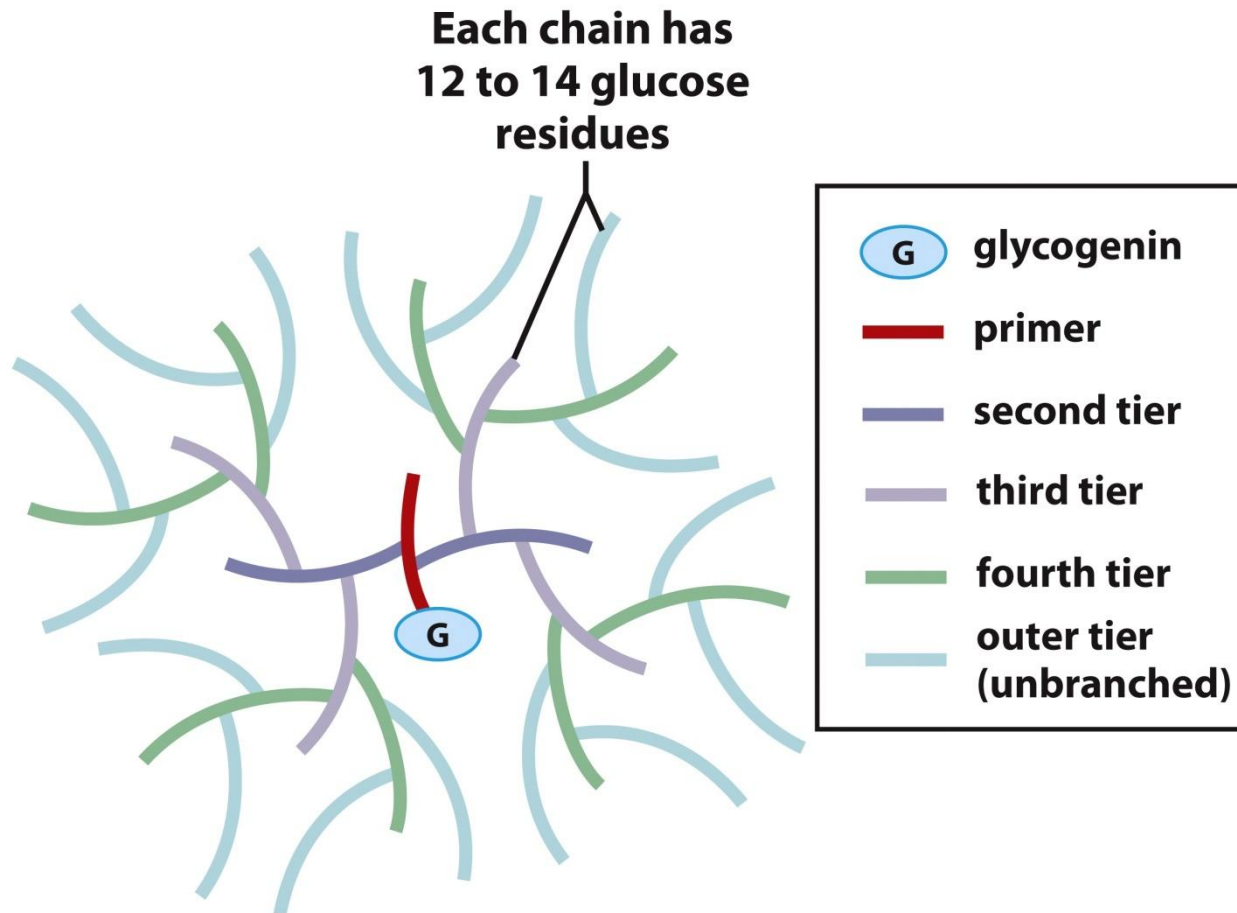


glycogenin



Glycogenesis (5)

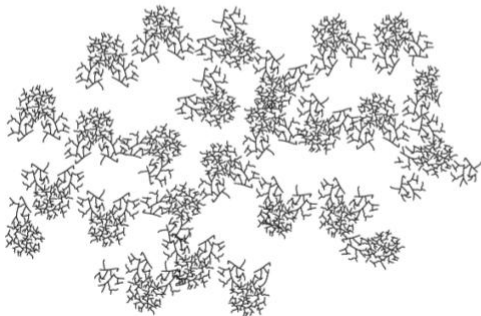
glycogenin and the structure of the glycogen particle



β particles

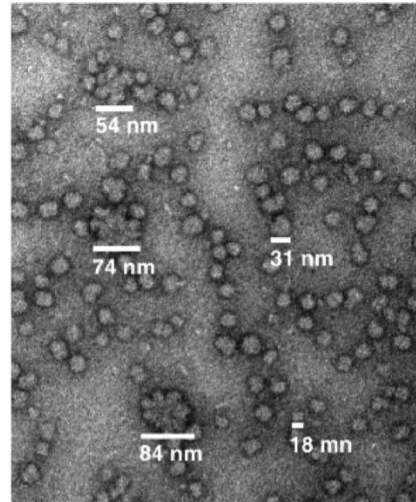


~ 20 nm

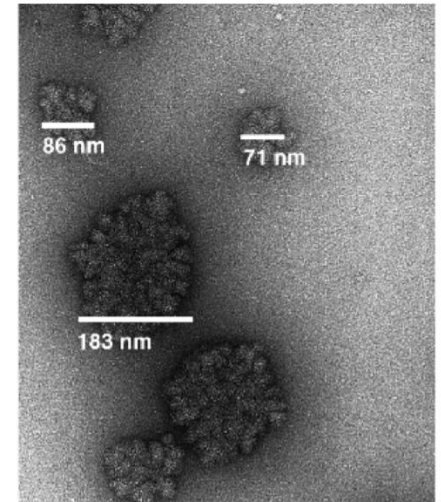


~ 100 nm

α particles



slipper limpet



rat liver



집신고둥 (바다 달팽이)

Summary

- (1) Glycogen breakdown is catalyzed by glycogen phosphorylase
- (2) G1P can enter glycolysis or replenish blood glucose in liver
- (3) The sugar nucleotide UDP-glucose donates glucose for glycogen synthesis
- (4) Glycogenin primes the initial sugar residues in glycogen

송길영 바이브컴퍼니 부사장

- (1) 당신은 혼자 삽니다 – 혼밥 혼영 등
- (2) 당신은 오래 삽니다 – 노후의 복지 문제, 세대갈등
- (3) 당신 없이도 사람들은 잘 삽니다 – 무인 시스템 선호

건강한 개인주의 vs 선택적 대면 전화 공포증

전화 매뉴얼 : 먼저 인사 (자기소개), 통화 괜찮은지 물어봄, 어떻게 지내는지 물어봄. 본론 얘기. 결론 마무리. 인사 후 끊기

정중히 거절하는 법, 답변을 유보하는 법, 대안을 제시하는 법
“나는 당신에게 호의를 가지고 있습니다” 는 마음

Today, in this class...



Regulation of
glycogen synthesis
& breakdown



Regulation of
metabolic
pathways

15.5 Regulation of glycogen synthesis and breakdown

insulin



glucagon



epinephrine



Liver / Muscle

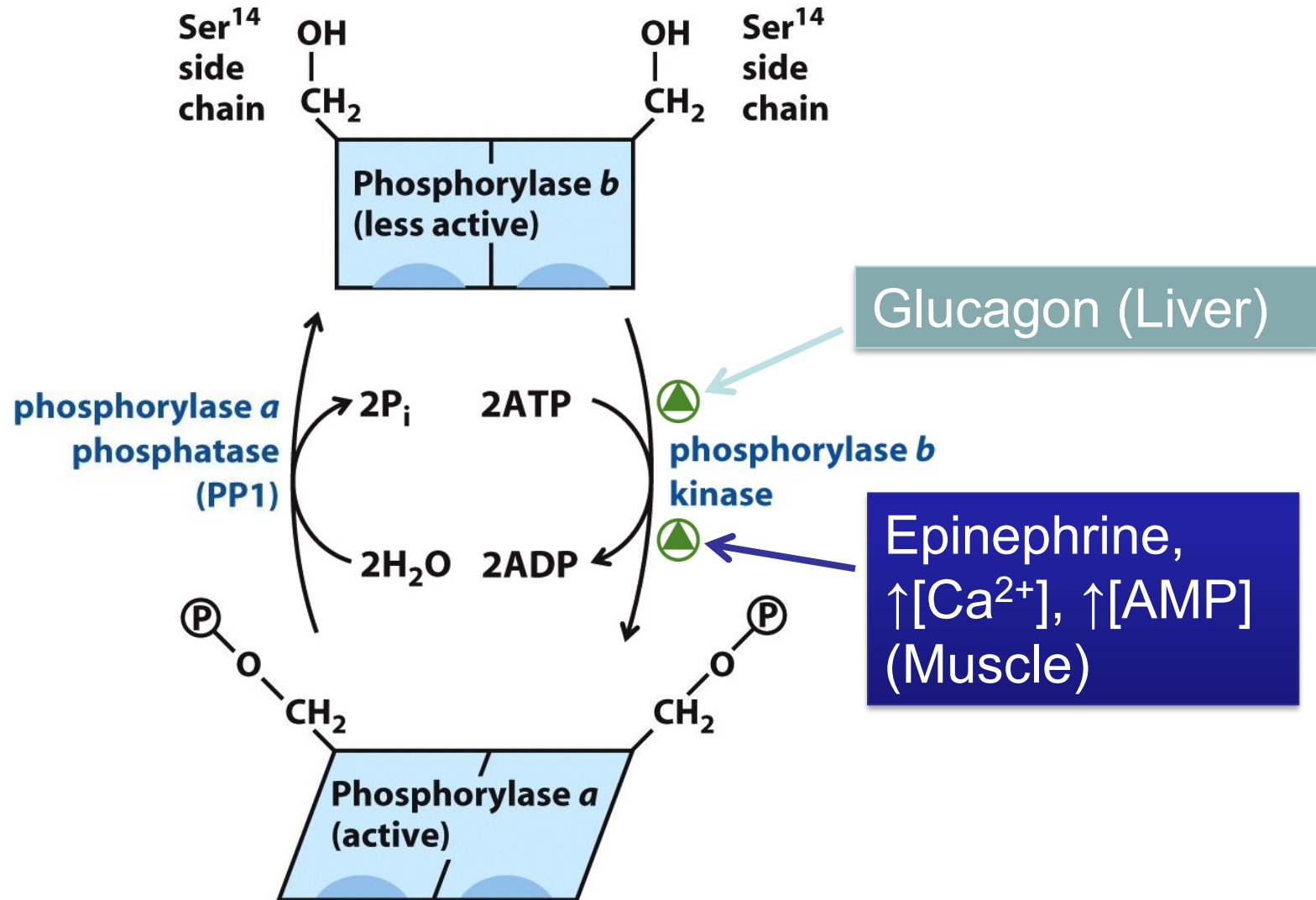
	TM G		G needed		E needed	
	insulin		glucagon		epinephrine	
	Liver	muscle	liver	muscle	liver	muscle
Glycolysis	up	up	down	-	down	up
Gluconeogenesis	down	-	up	-	up	-
Glycogenesis	up	up	down	-	down	down
Glycogenolysis	down	down	up	-	up	up

Muscle lacks receptors for glucagon

Muscle lacks enzymes for gluconeogenesis

Glycogenolysis (1) : by glucagon and epinephrine

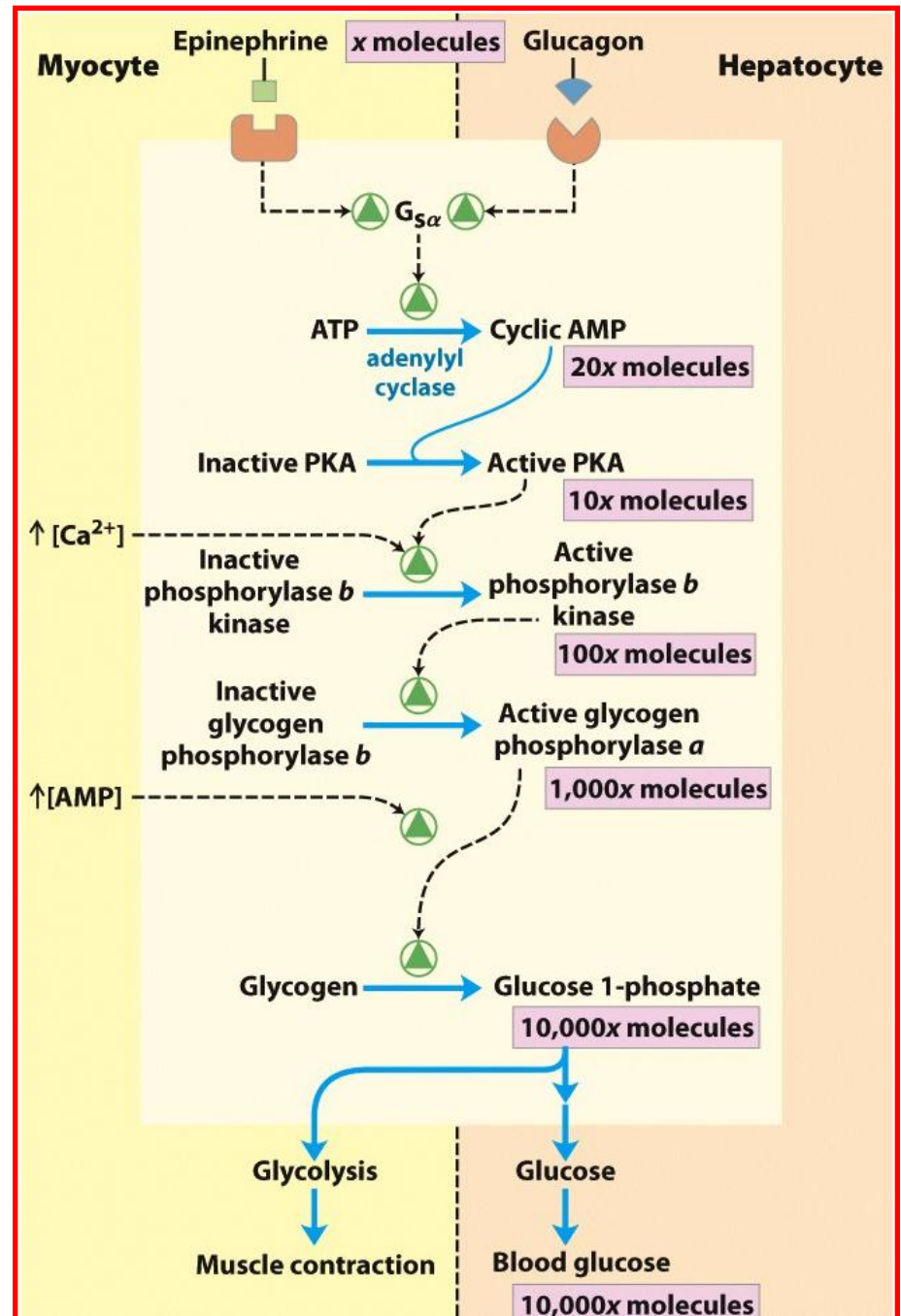
activation of glycogen phosphorylase by phosphorylation



Glycogenolysis (1)

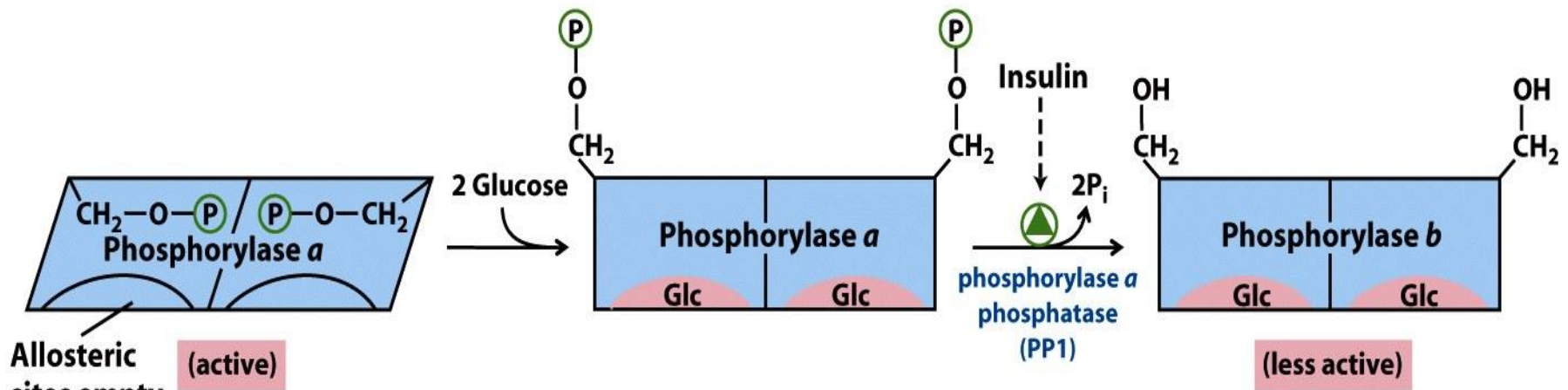
by Glucagon & EP

: glycogen phosphorylase
enzyme cascade by
epinephrine or glucagon



Glycogenolysis (2) : inhibited by glucose

glycogen phosphorylase is inactivated by the **binding of glucose** followed by the **dephosphorylation**



Allosteric
sites empty

(active)

Figure 15-36

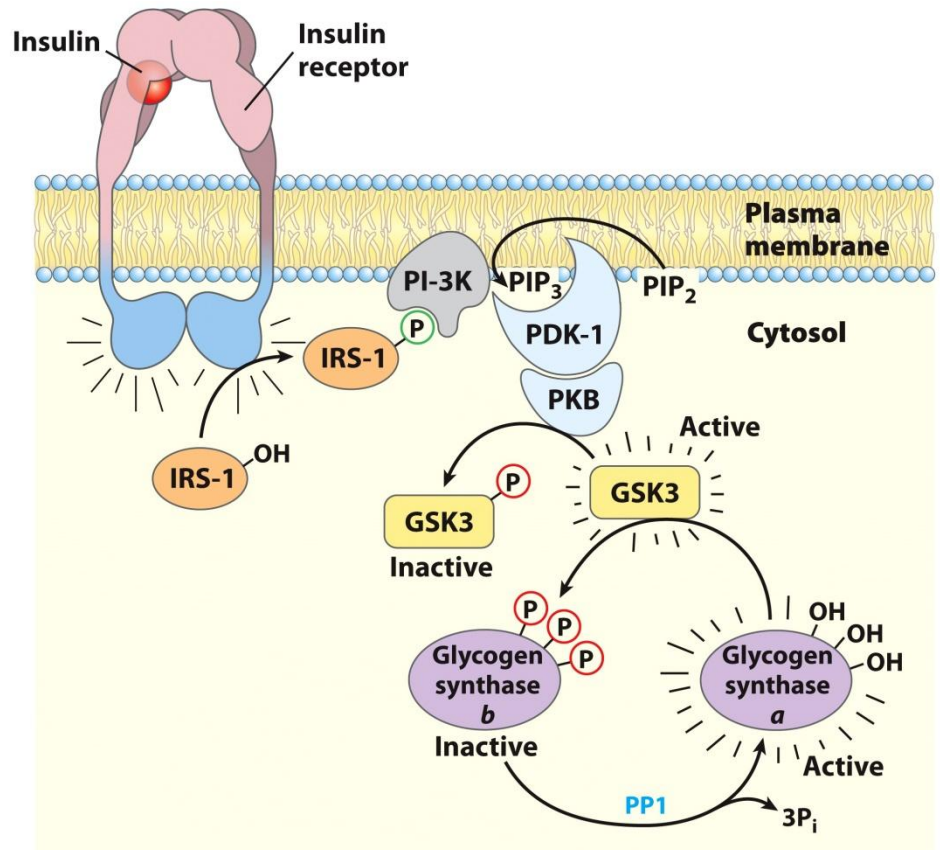
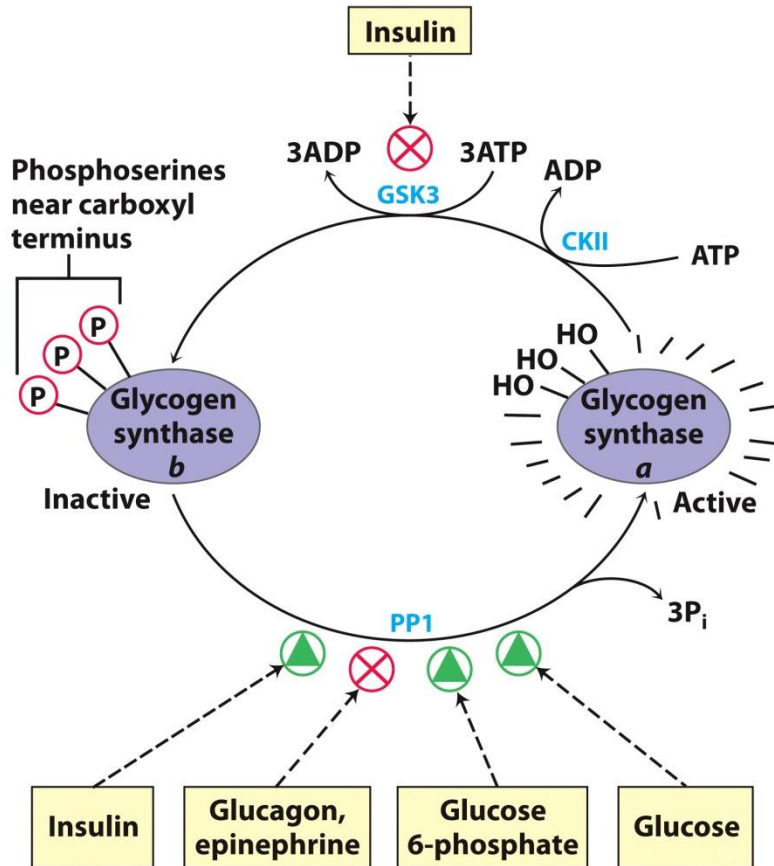
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Glycogenesis (1) : insulin

phosphorylation of glycogen synthase (by **GSK3**) is suppressed

→ Activation of GS

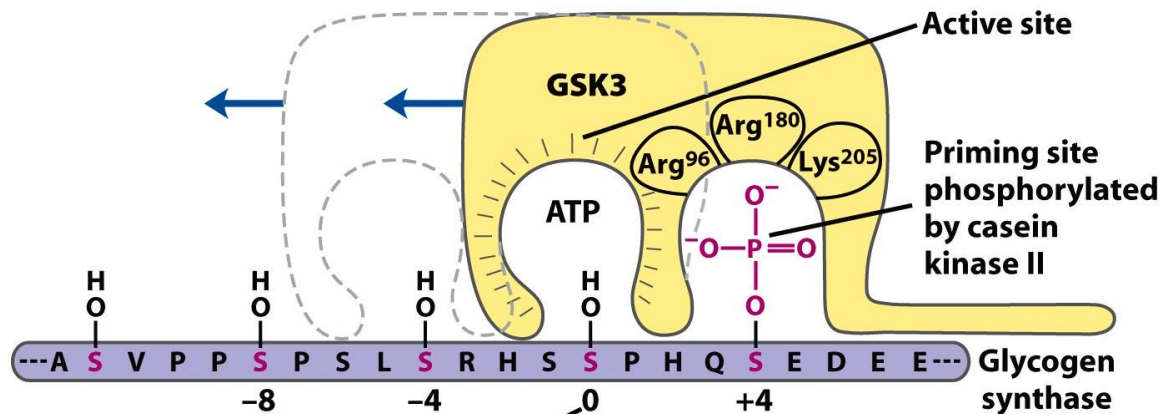


GSK3: glycogen synthase kinase 3

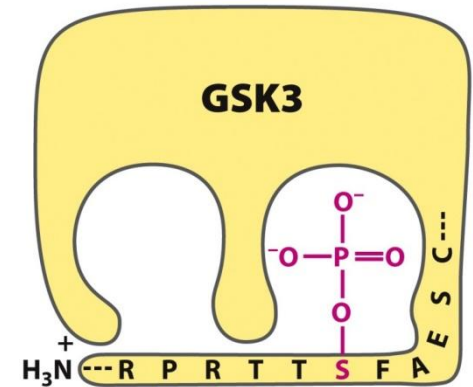
Glycogenesis (2) : GS priming by CKII

- priming on GS by casein kinase II (CKII) → pGS up → GS inactive
- pseudosubstrate of GSK3 inhibits GSK3 → pGS down → GS active

(activated)



(inactivate)



P

PKA
PKB

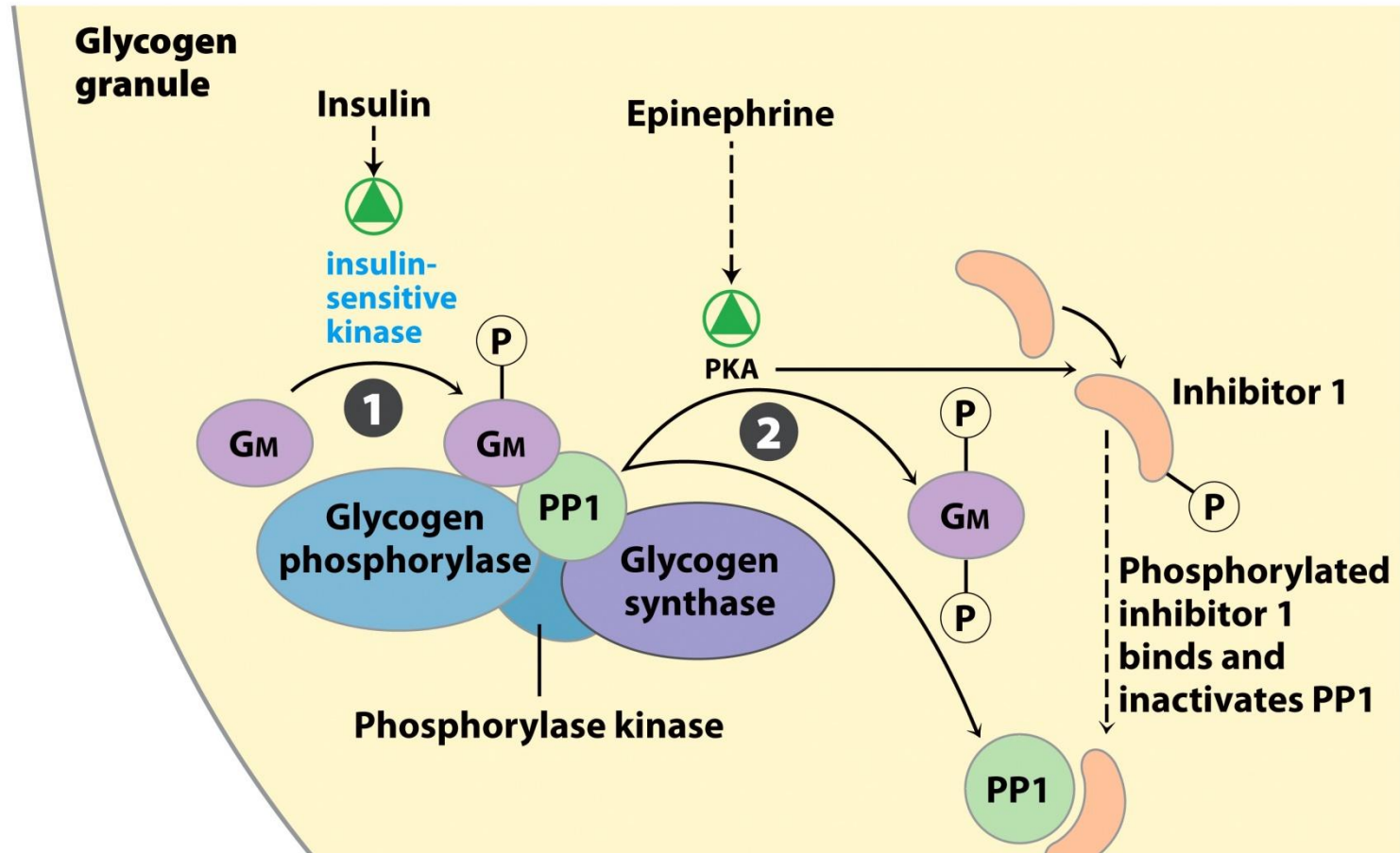
(inactivate)

P

PP1

(activate)

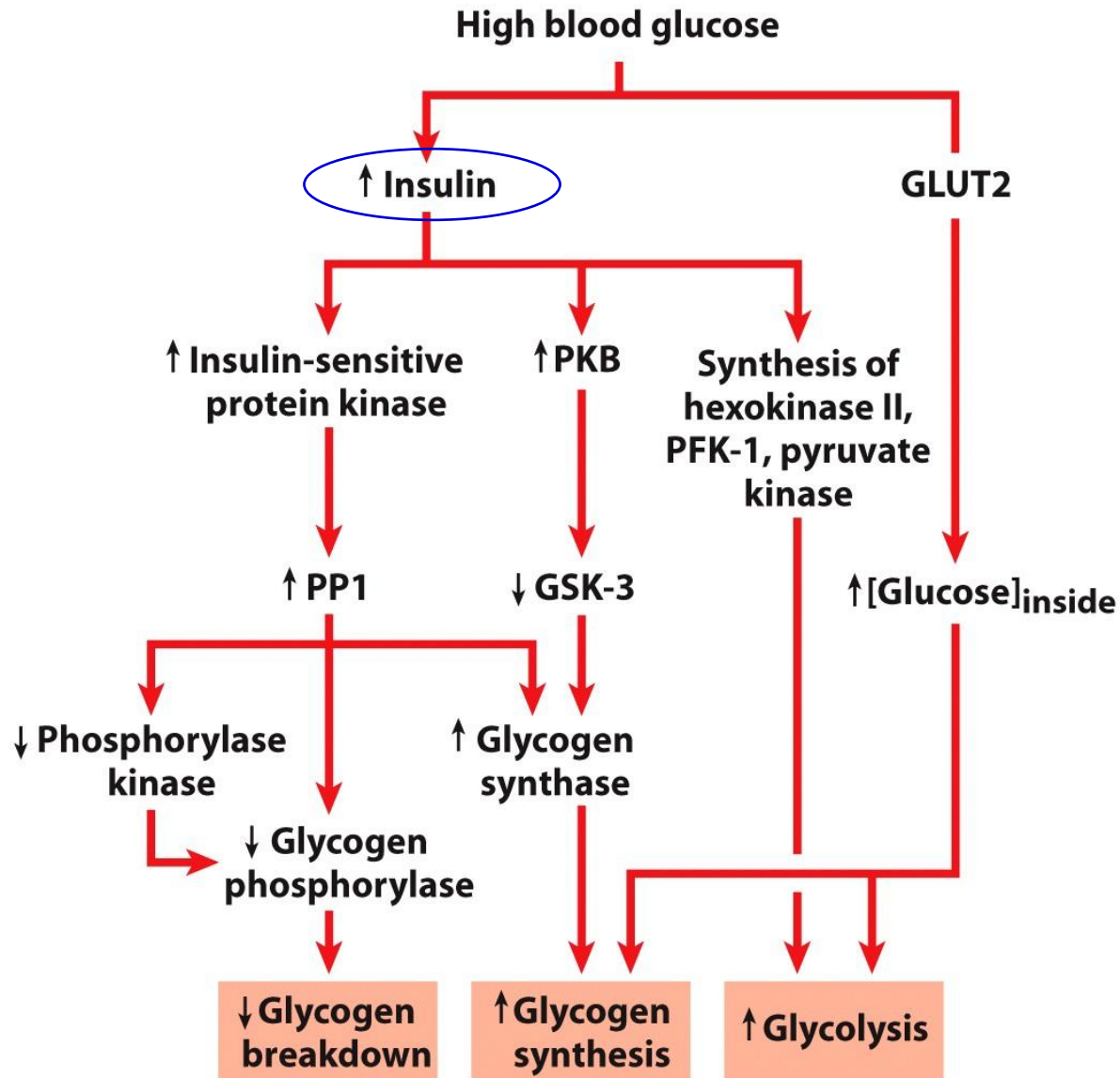
Glycogenesis or Glycogenolysis : regulation by insulin / EP through **glycogen-targeting protein GM**



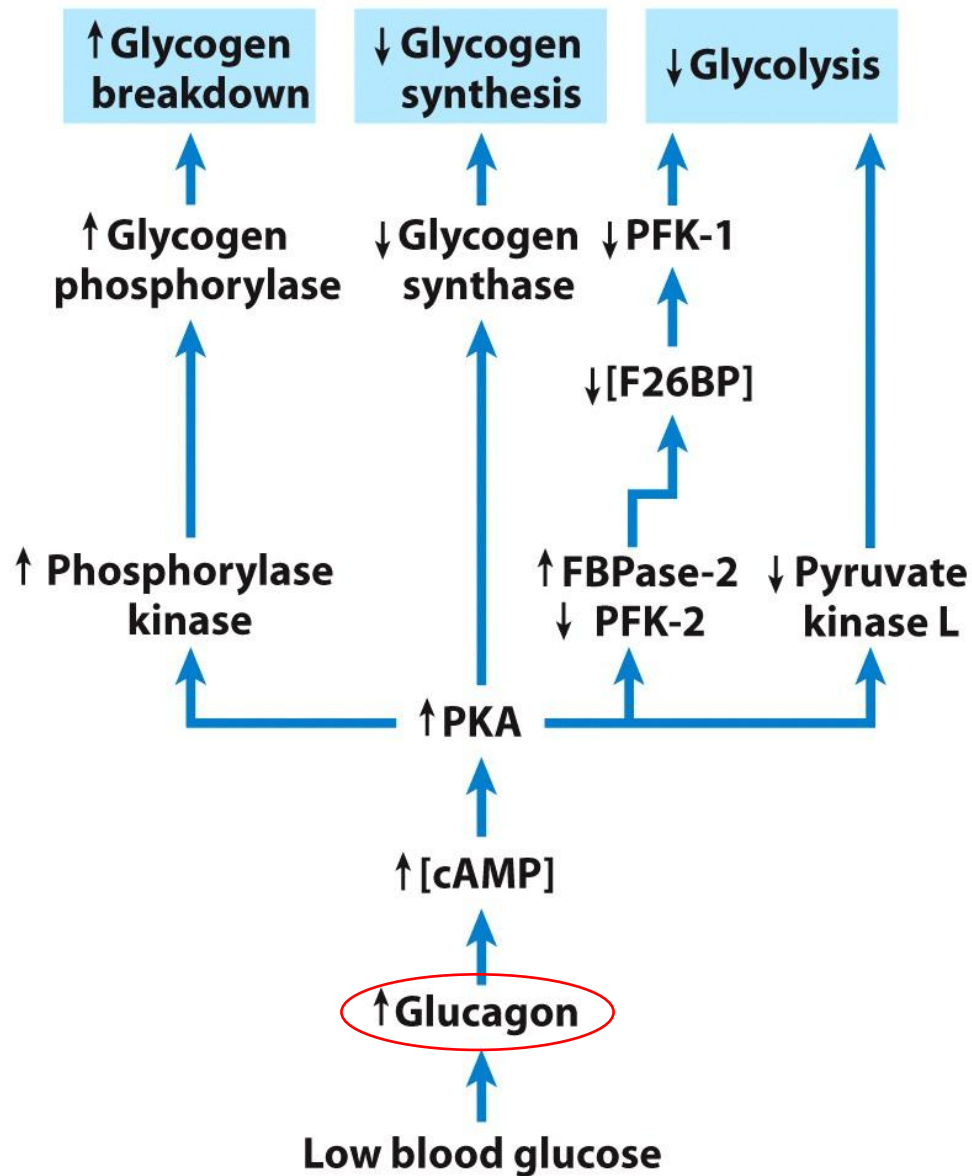
Insulin → **GM site 1 P** → PP1 act → deP → glycogenesis

Epinephrine → **GM site 2 P** → PP1 inact → P → glycogenolysis

Regulation of carbohydrate metabolism in the **liver**



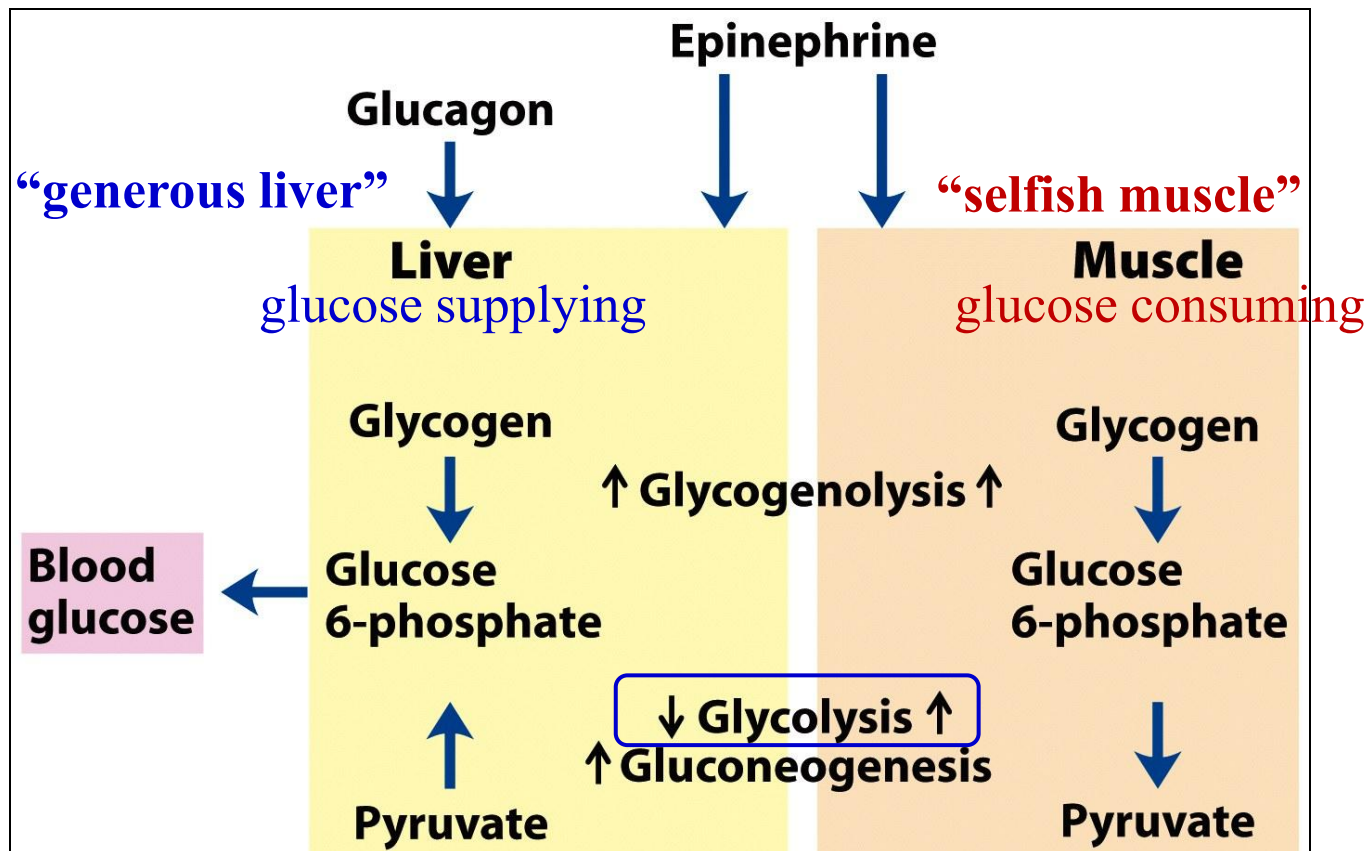
Regulation of carbohydrate metabolism in the **liver**



Difference in the regulation of carbohydrate metabolism in **liver** and **muscle**

(1) **Muscle** lacks enzymes for gluconeogenesis

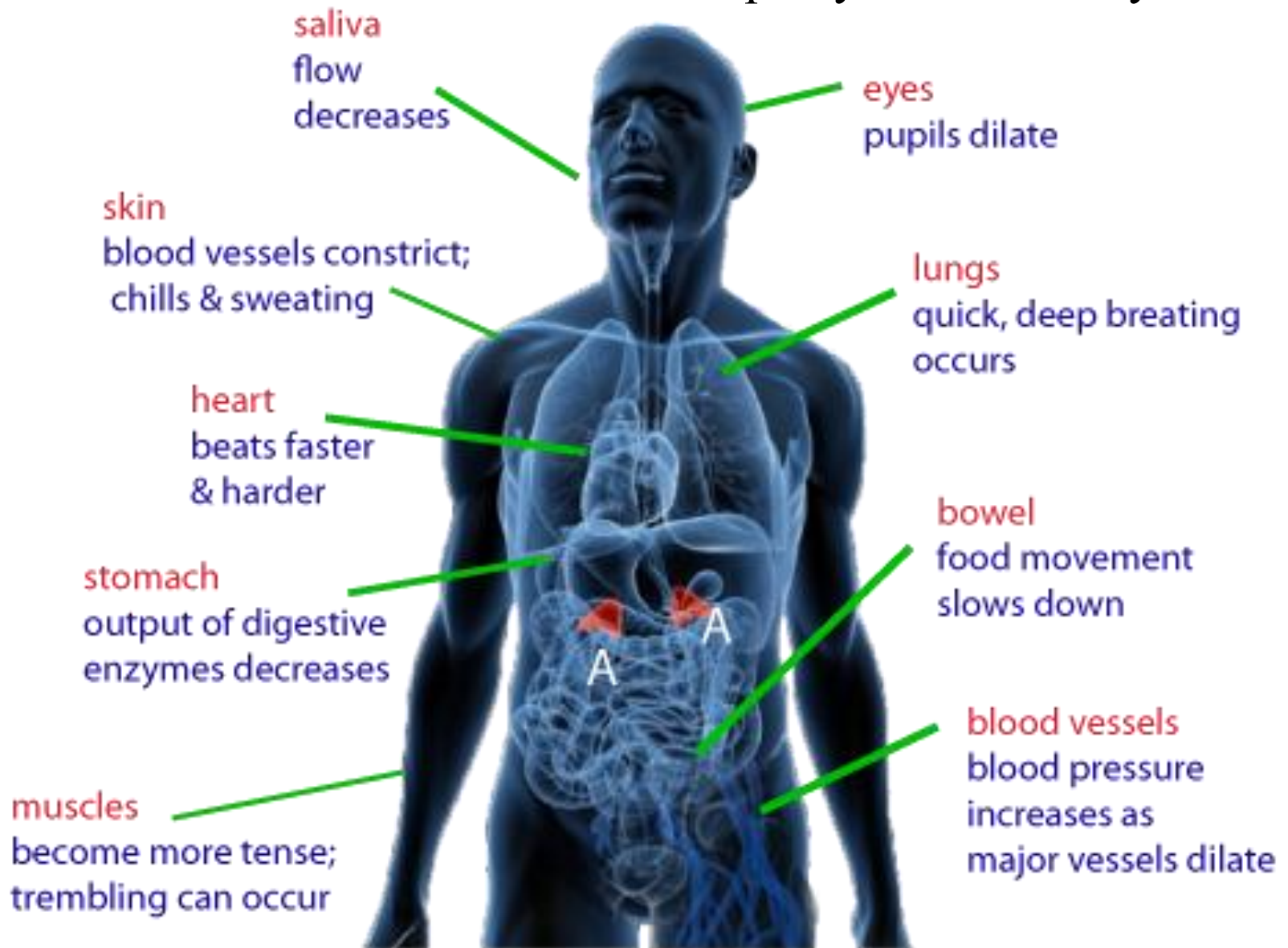
(2) **Muscle** lacks receptors for glucagon



→ Muscle consumes glucose to prepare energy for fight-or-flight response

Fight or Flight Response

<https://youtu.be/m2GywoS77qc>

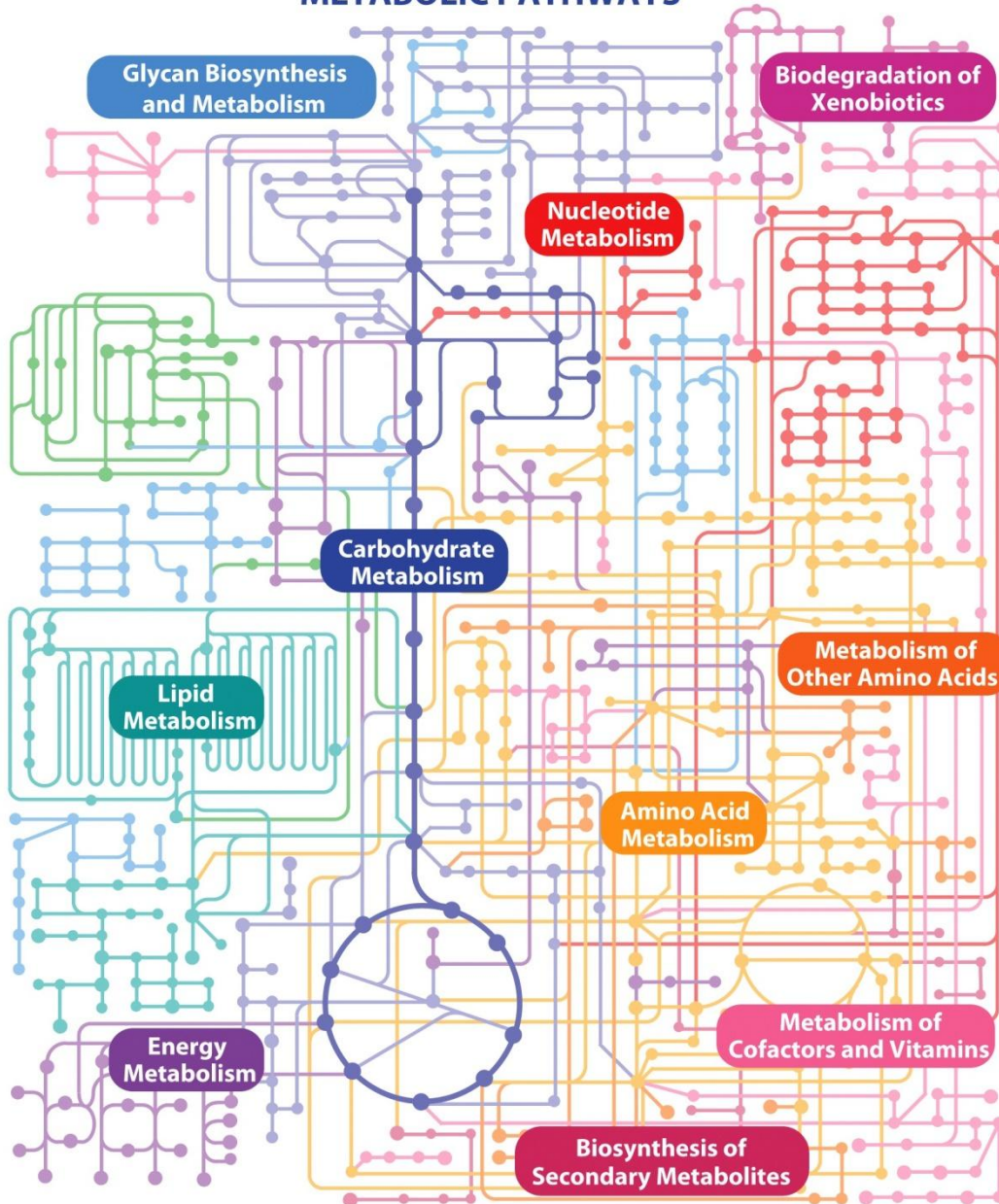


Summary

- (1) Glycogen phosphorylase is regulated allosterically and hormonally thru P and deP
- (2) Glycogen synthase is also regulated by P and deP
- (3) GSK3 mediates insulin action
- (4) PP1 is central to glycogen metabolism
- (5) Liver: glucagon → glycogenolysis and gluconeogenesis to supply glucose for other tissues
- (6) Muscle: epinephrine → glycogenolysis and glycolysis to provide ATP

15.1 Regulation of metabolic pathways

METABOLIC PATHWAYS



Dynamic or Steady?

~ 30,000 proteins
thousands of reactions
hundreds of metabolites

→ dynamic steady state
(homeostasis)

Dynamic Steady State

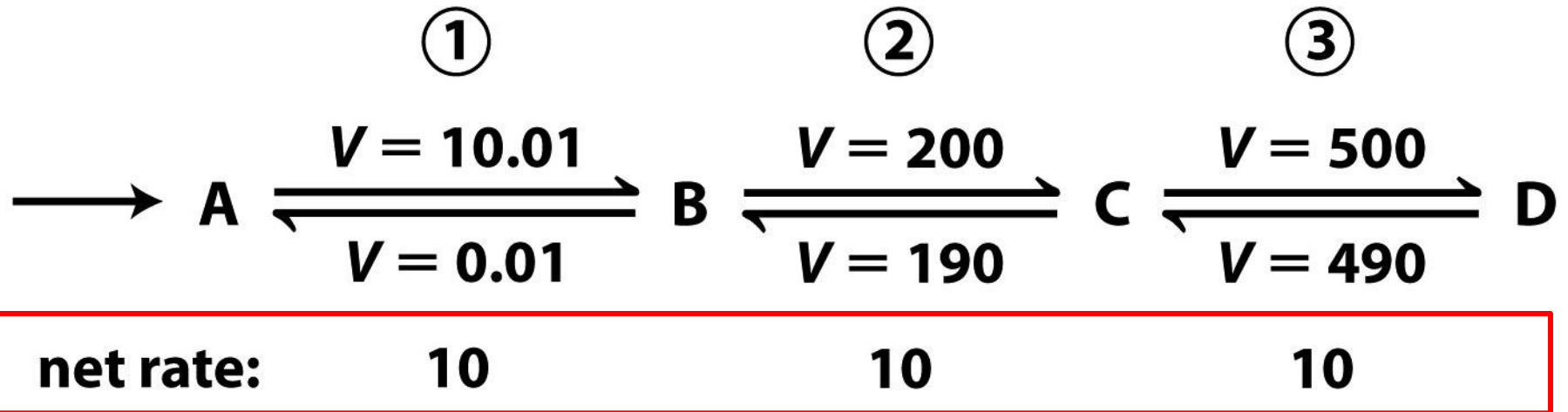
- 1) Fuels enter a cell and waste leave, but **the gross composition of a cell** or an organism **do not change** appreciably over time.
- 2) Dynamic steady state
 - Maintaining **homeostasis**: constant substrate conc.
 - Metabolic flow (**flux**):



$$\mathcal{V}_1 = \mathcal{V}_2 \longrightarrow \mathbf{Constant [S]}$$

- ex. [Glc] in blood: 5 mM (homeostasis)

Reactions far from Eq. in cells are common points of regulation



steps (2) and (3) are near equilibrium,

: $\Delta G'$, is close to zero. An increase in $[C]$ or $[D]$ can reverse the direction of these steps.

step (1) is maintained in the cell far from equilibrium

: forward rate greatly exceeds its reverse rate.

: being subject to regulatory mechanism

Reactions far from equilibrium : hexokinase, PFK-1, pyruvate kinase, pyruvate carboxylase

TABLE 15-3

Equilibrium Constants, Mass Action Coefficients, and Free-Energy Changes for Enzymes of Carbohydrate Metabolism

Enzyme	K'_{eq}	Mass action ratio, Q		Reaction near equilibrium in vivo?*	$\Delta G'^{\circ}$ (kJ/mol)	$\Delta G'$ (kJ/mol) in heart
		Liver	Heart			
Hexokinase	1×10^3	2×10^{-2}	8×10^{-2}	No	-17	-27
PFK-1	1.0×10^3	9×10^{-2}	3×10^{-2}	No	-14	-23
Aldolase	1.0×10^{-4}	1.2×10^{-6}	9×10^{-6}	Yes	+24	-6.0
Triose phosphate isomerase	4×10^{-2}	—	2.4×10^{-1}	Yes	+7.5	+3.8
Glyceraldehyde 3-phosphate dehydrogenase + phosphoglycerate kinase	2×10^3	6×10^2	9.0	Yes	-13	+3.5
Phosphoglycerate mutase	1×10^{-1}	1×10^{-1}	1.2×10^{-1}	Yes	+4.4	+0.6
Enolase	3	2.9	1.4	Yes	-3.2	-0.5
Pyruvate kinase	2×10^4	7×10^{-1}	40	No	-31	-17
Phosphoglucose isomerase	4×10^{-1}	3.1×10^{-1}	2.4×10^{-1}	Yes	+2.2	-1.4
Pyruvate carboxylase + PEP carboxykinase	7	1×10^{-3}	—	No	-5.0	-23
Glucose 6-phosphatase	8.5×10^2	1.2×10^2	—	Yes	-17	-5.0

Source: K'_{eq} and Q from Newsholme, E.A. & Start, C. (1973) *Regulation in Metabolism*, Wiley Press, New York, pp. 97, 263. $\Delta G'$ and $\Delta G'^{\circ}$ were calculated from these data.

*For simplicity, any reaction for which the absolute value of the calculated $\Delta G'$ is less than 6 is considered near equilibrium.

Regulation of enzyme activities

(1) **Amount:** number of enzyme molecules

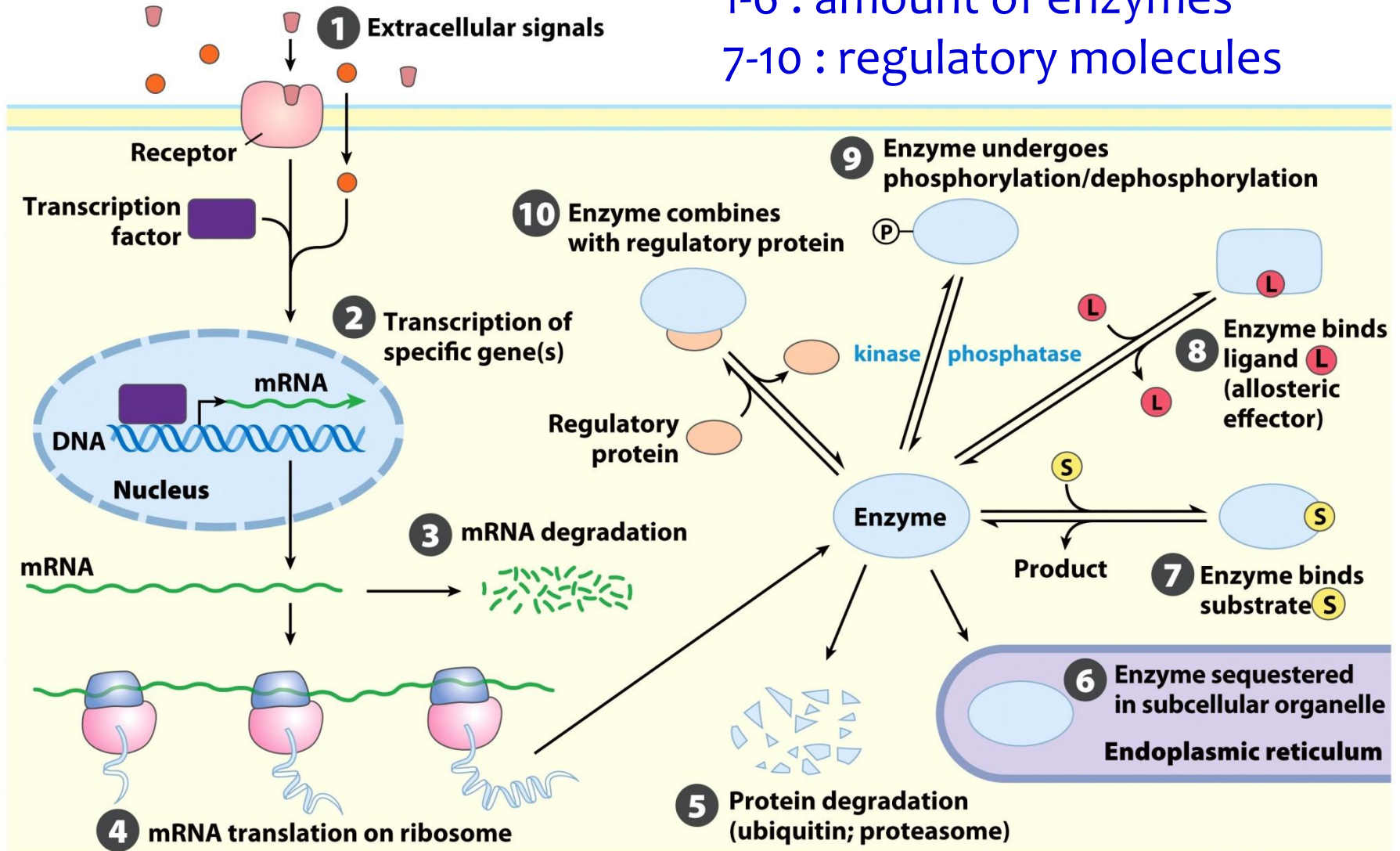
(2) **Catalytic activity:** milliseconds to hours time scale

- **Changes in local concentration of small molecules**
 - substrate (G for glycolysis)
 - product (ATP from glycolysis)
 - key metabolite or cofactor (NADH)
- **Second messengers (cAMP, Ca^{2+})**
 - mediates allosteric regulation
- **Rates of synthesis and degradation of enzymes**
 - Transcription factors & response elements

Factors affecting the activity of enzymes

1-6 : amount of enzymes

7-10 : regulatory molecules



ATP and AMP are key cellular regulators

- A 10% decrease in [ATP] can greatly affect the activity of ATP utilizing enzymes
- A 10% decrease in [ATP] leads to a dramatic increase in [AMP]
 - AMP can be a more potent allosteric regulator

TABLE 15–4 Relative Changes in [ATP] and [AMP] When ATP Is Consumed

Adenine nucleotide	Concentration before ATP depletion (mM)	Concentration after ATP depletion (mM)	Relative change
ATP	5.0	4.5	10%
ADP	1.0	1.0	0
AMP	0.1	0.6	600%

Adenylate kinase and AMPK works for maintaining [ATP]/[AMP] ratio

Adenylate kinase produces AMP from ADP

$2\text{ADP} \rightarrow \text{AMP} + \text{ATP}$
(restoring ATP levels)

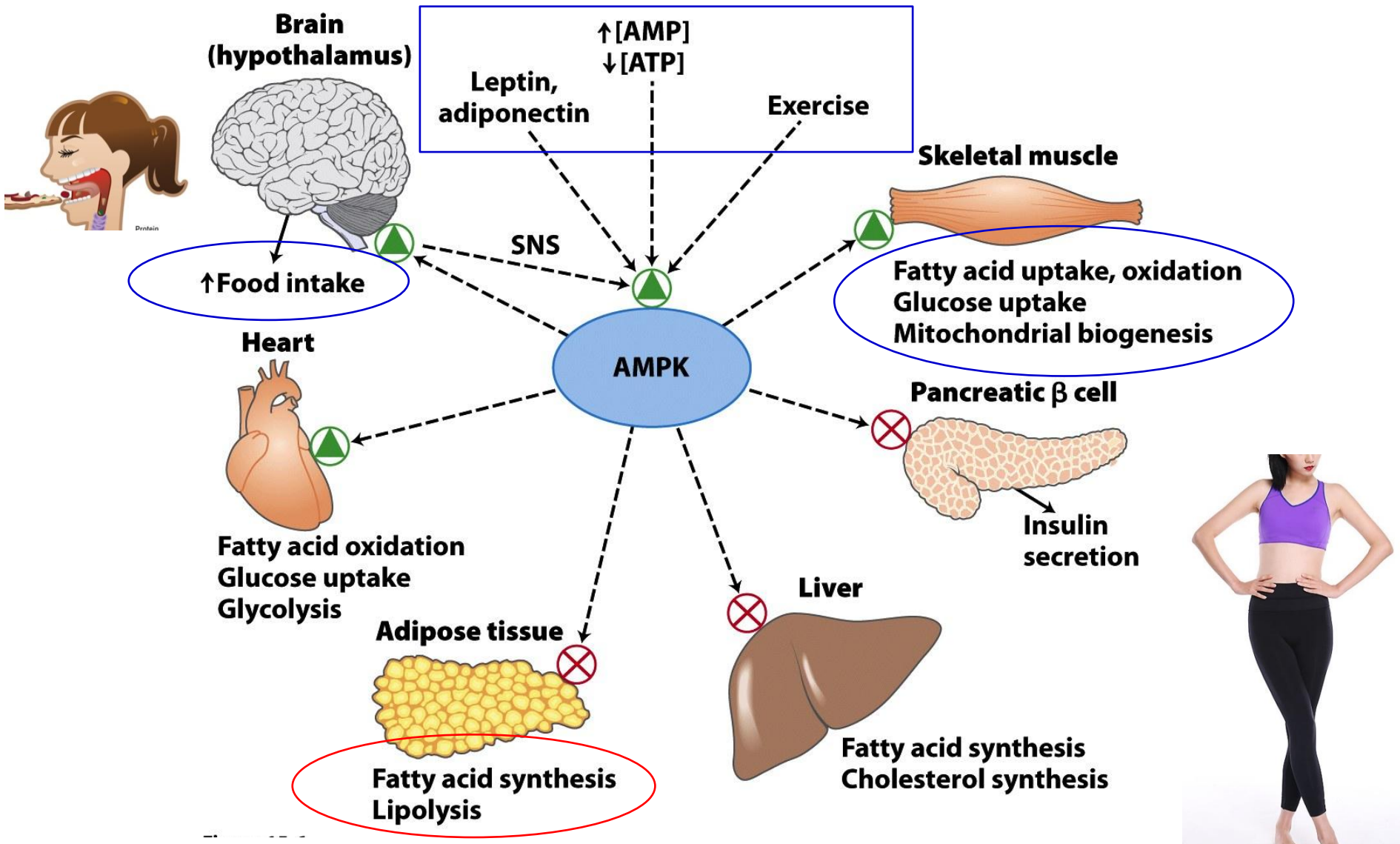
AMP-activated protein kinase (AMPK)

: responds to an increase in [AMP]

→ Increase glucose transport, activate glycolysis, and fatty acid oxidation

→ To raise [ATP] and lower [AMP]

Role of AMP-activated protein kinase (AMPK) in carbohydrate and fat metabolism

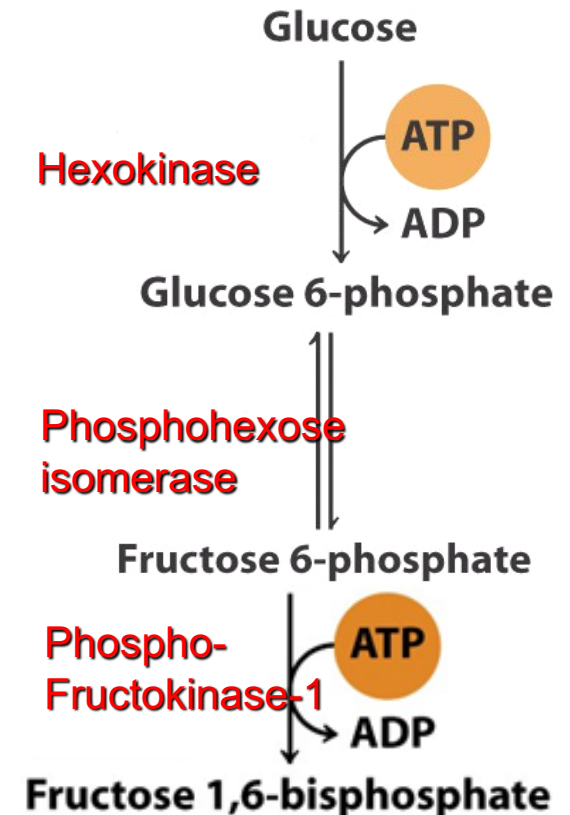
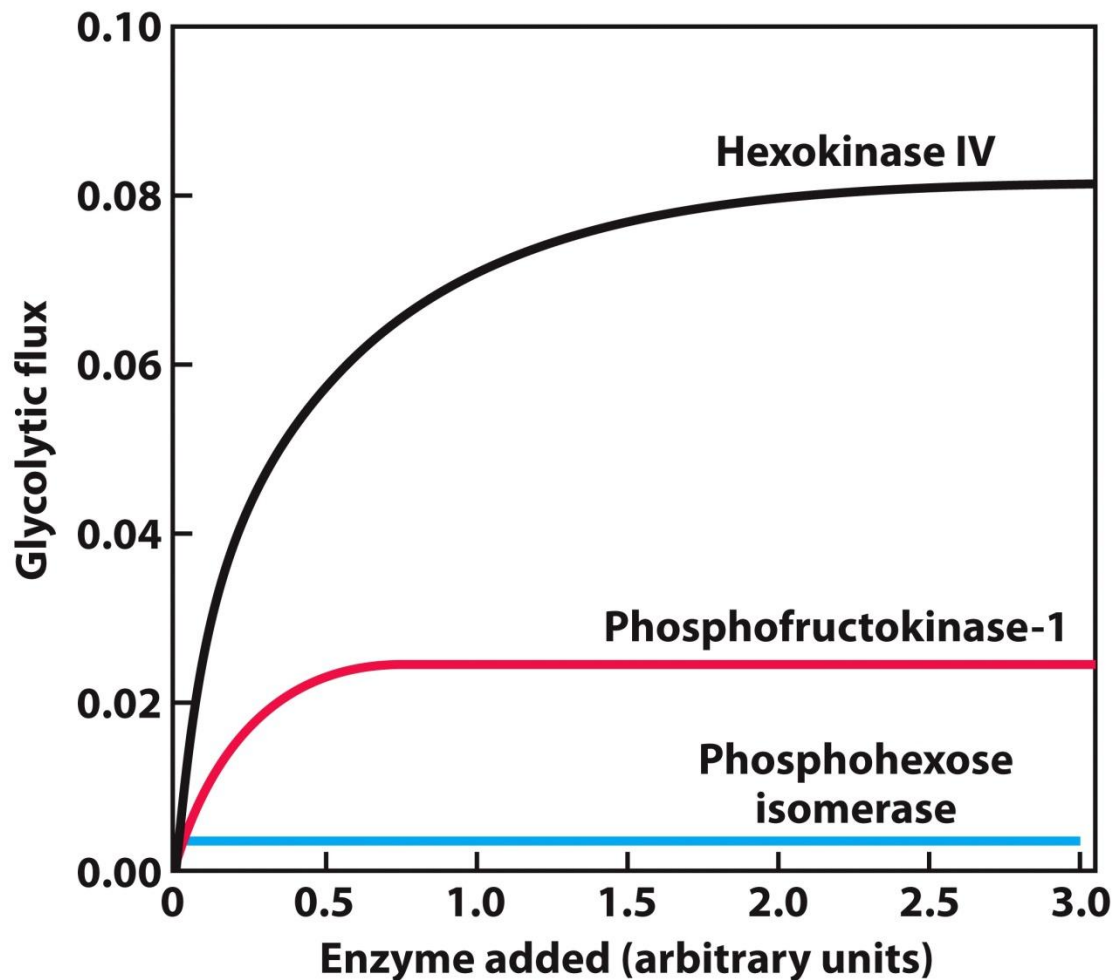


Summary

1. Compensating changes in enzymatic activities for maintaining dynamic steady state
2. 10 factors for regulation of enzyme activities
3. In multistep processes : reactions far from Eq. are the points of regulation
4. Regulatory mechanisms maintain nearly constant level of [ATP], [NADH], and [blood glucose]
5. Adenylate kinase and AMPK works for maintaining [ATP]/[AMP] ratio

15.2 Analysis of Metabolic Control

1) Contribution of each enzyme to flux



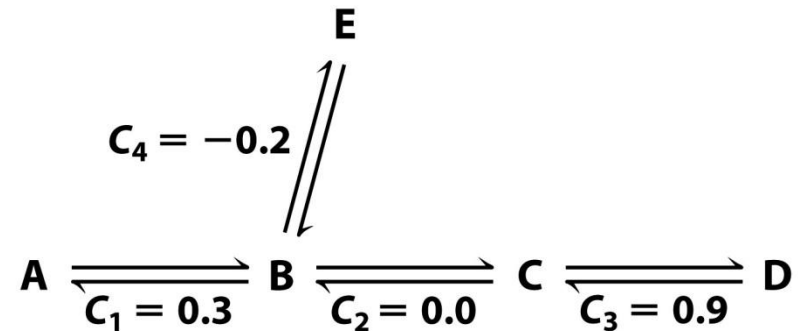
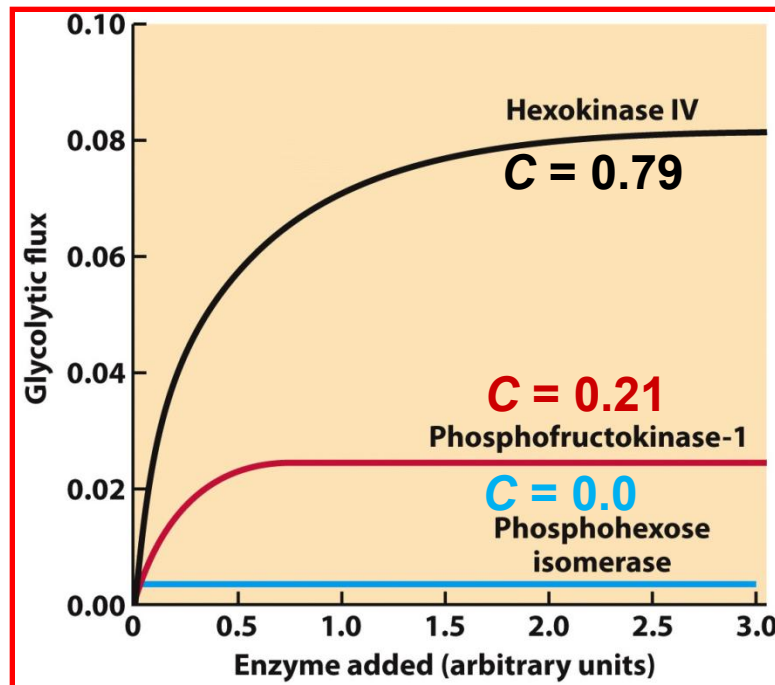
2) Flux Control Coefficient, C

- Quantify the effect of enz activity change on metabolic flux thru a pathway
- 0.0 – 1.0 scale, sum is 1.0

0.0: an enz with no impact on the flux

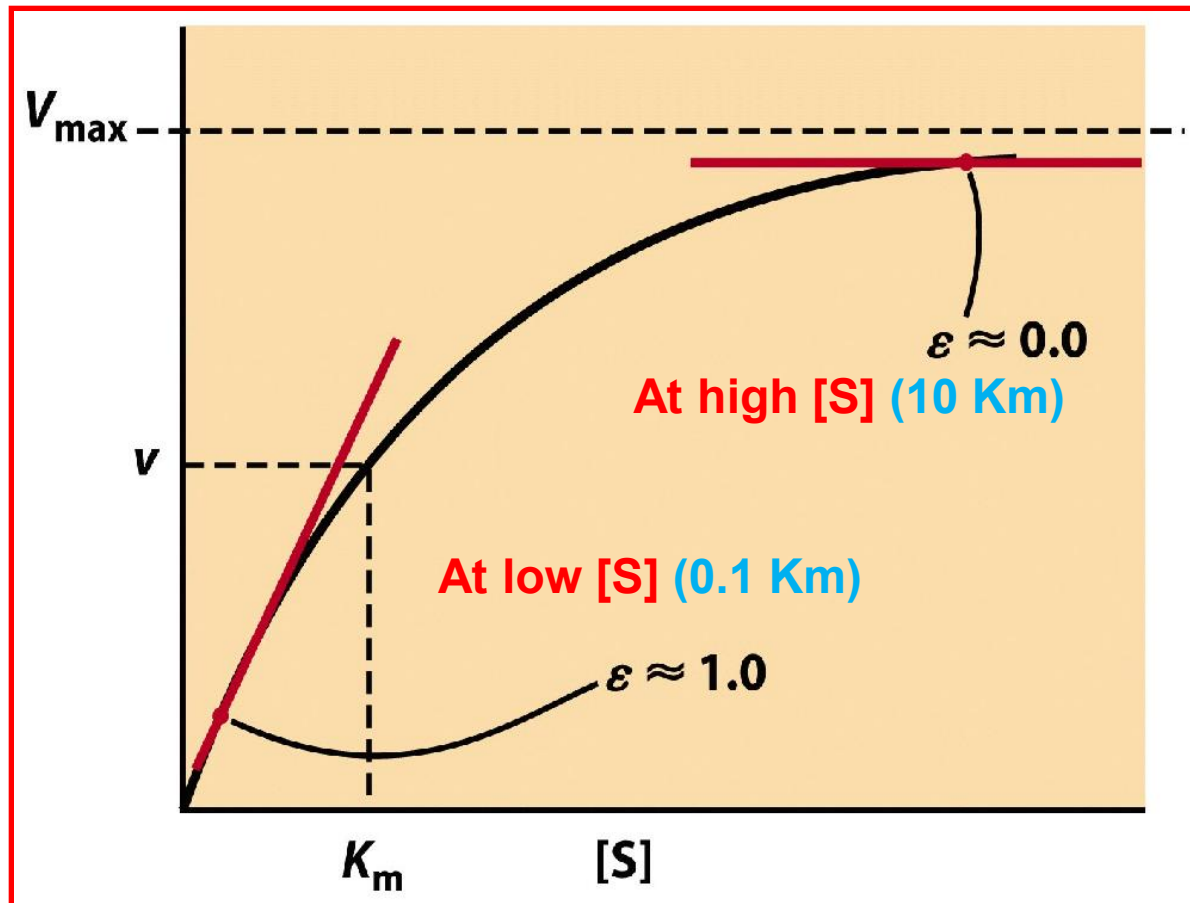
1.0: an enz that wholly determines the flux

An enz can also have negative value



3) Elasticity Coefficient (ϵ)

- Enz's responsiveness to changes in [Metabolite] or [Regulator]



4) Response coefficient (R)

- Effect of outside controller on flux
- Not by metabolite nor enz
- e.g. insulin as an outside controller

$$R = C \cdot \varepsilon$$

Response coefficient R is a function of :

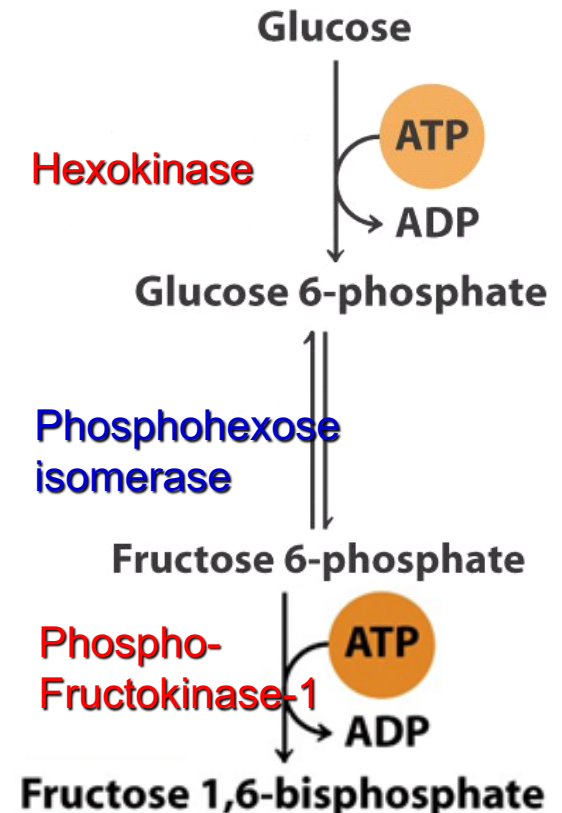
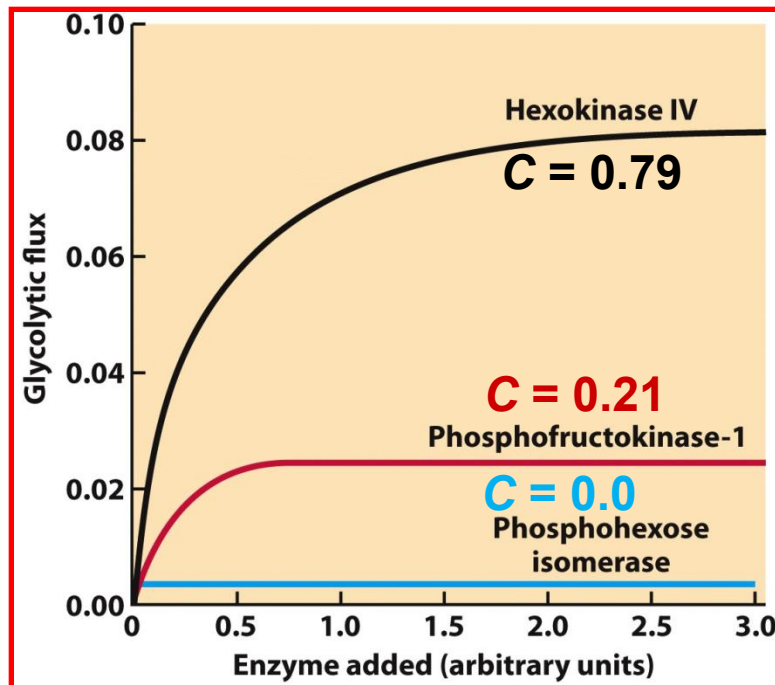
- (1) How sensitive the pathway is to change in the activity of enzyme (control coefficient, C)
- (2) How sensitive that specific enzyme is to change in the outside controlling factor (elasticity, ε)

5) Enzyme for flux control

5 x increase of PFK-1

→ < 10% effect on glycolysis

→ Hexokinase, not PFK-1, is most influential in glycolysis.



Control or Regulation ?

(control the flux)

(rebalance the level of metabolites)

Role of hexokinase

: actually alter the flux through the pathway

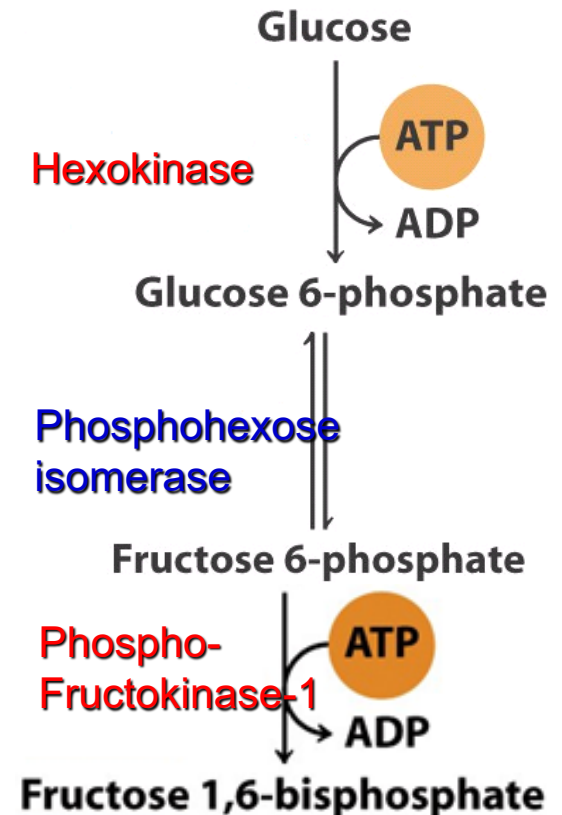
→ **Control mechanism**

Role of PFK-1

: act to maintain metabolite concentrations

: metabolite homeostasis – to prevent large changes in metabolite concentrations

→ **Regulatory mechanism**



Summary

(1) Metabolic Control Analysis

: control of metabolic flux is distributed to enzymes in the path

(2) Flux control coeff

: the effect of an enzyme's concentration on flux

(3) Elasticity coeff

: responsiveness to changes in [metabolite]

(4) Response coeff

: measure of change in flux in response to an outside factor

: $R = C \cdot \varepsilon$

(5) Control (control the flux) **& Regulation** (rebalance the level of metabolites)